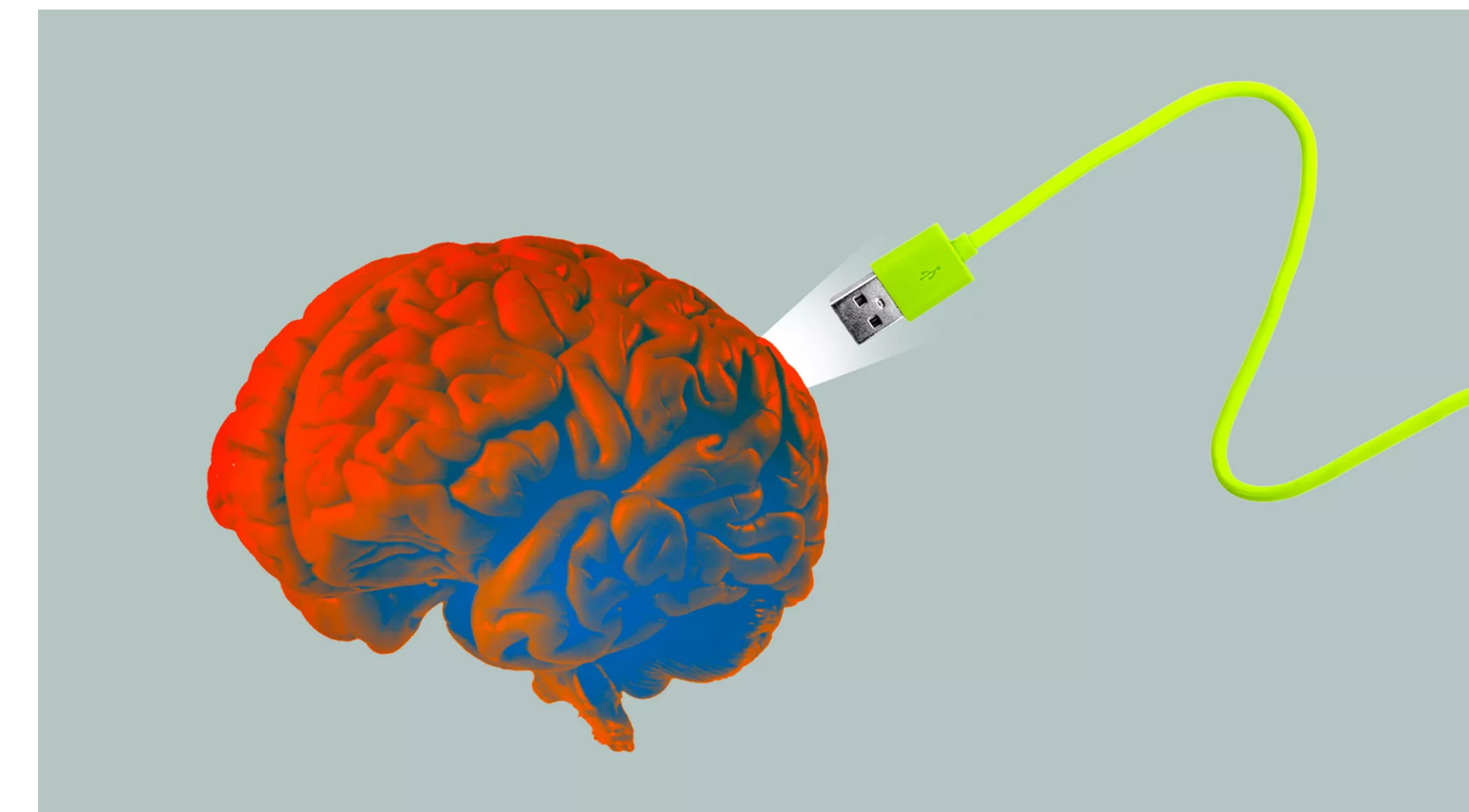
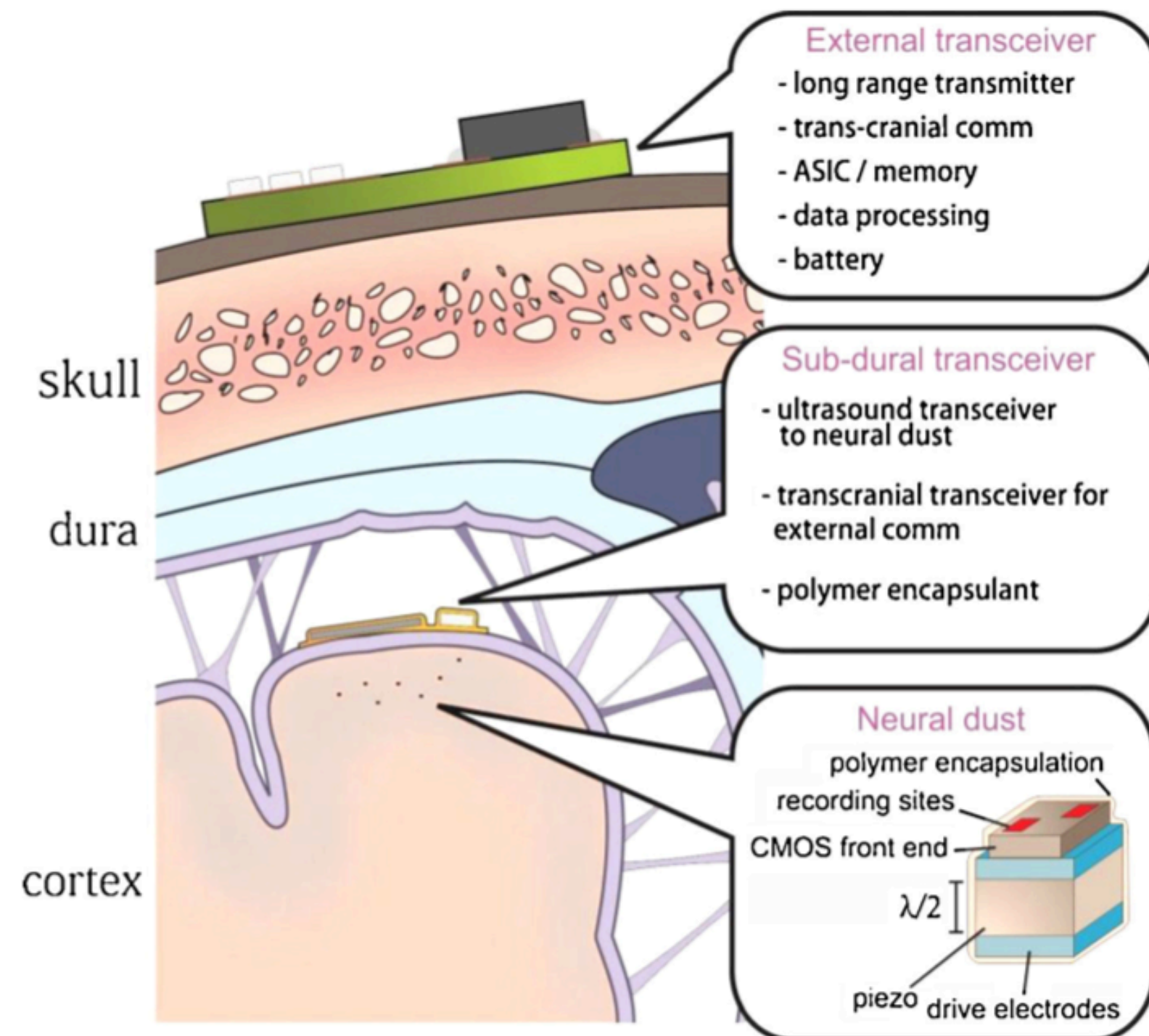


Neurotechnologies Reading Group

- Place to discuss emerging technologies for neuroscience!
- Understand fundamental principles of operation, trade-offs etc
- Big picture thinking: what problems might these technologies solve in ~10 year period?
- What are the fundamental bottlenecks with existing tools?
- Discussions + crazy ideas... collaborations?



Neural Dust



Neural Dust: An Ultrasonic, Low Power Solution for Chronic Brain-Machine Interfaces

Dongjin Seo ^{*}, Jose M. Carmena ^{*} [†], Jan M. Rabaey ^{*}, Elad Alon ^{*} [‡], and Michel M. Maharbiz ^{*} [‡]

^{*}Department of Electrical Engineering and Computer Sciences and [†]Helen Wills Neuroscience Institute, University of California, Berkeley, CA, USA

[‡]Joint senior authors

Proposal: arXiv:1307.2196v1 (2013)

Validation: <http://dx.doi.org/10.1016/j.jneumeth.2014.07.025> (2015)

Recording: Seo et al., *Neuron* (2016)

Stimulation: Piech et al., *Nat. Bio. Eng.* (2020)

NEUROSCIENCE

Astellas to acquire neural dust start-up Iota Biosciences \$400M

by [Ryan Cross](#)

October 24, 2020 | A version of this story appeared in [Volume 98, Issue 41](#)

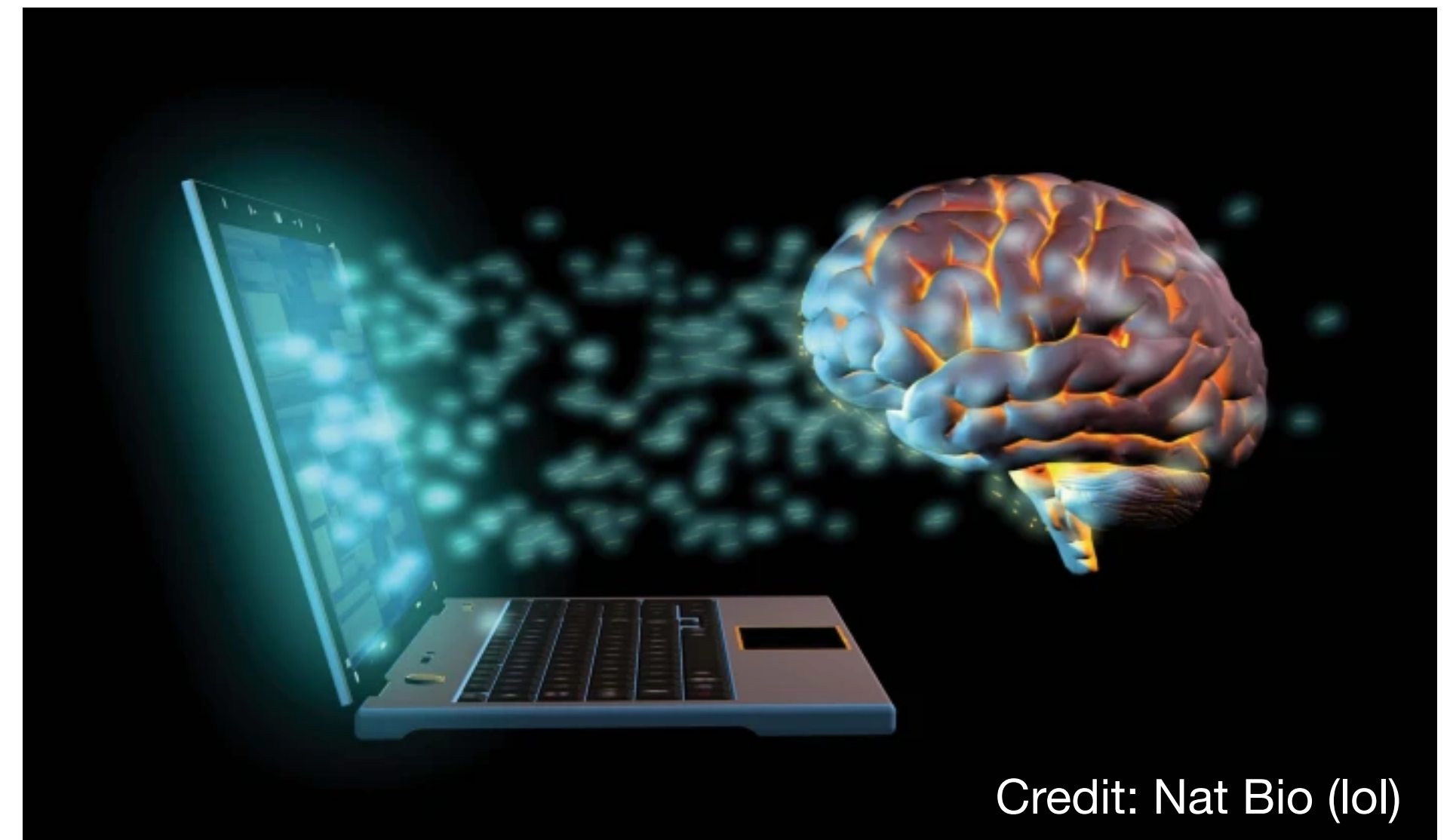
Jacques Carolan

Chronic Brain-Machine Interfaces

- BMI: “A device that translates neural information into commands capable of controlling external hardware of software”¹
- e.g. assisted living devices for individuals with motor or sensory impairments
- Responsive neurostimulation (RNS) for drug resistant epilepsy (primitive in their sensory modalities)²
- Augmented cognition (???)
- **Requirements:**
 - Wireless (c.f. N3 DARPA)
 - High-density (1000s)
 - Chronic (years)

¹ Smalley *Nature Biotechnology* (2019)

² Gummadaelli *Front. Neurosci.* (2018)



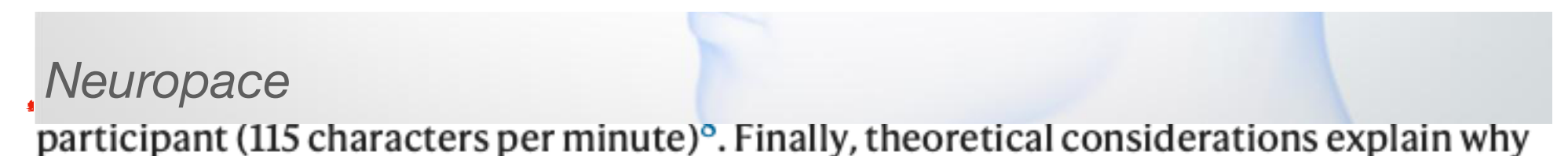
Article | Published: 12 May 2021



Health

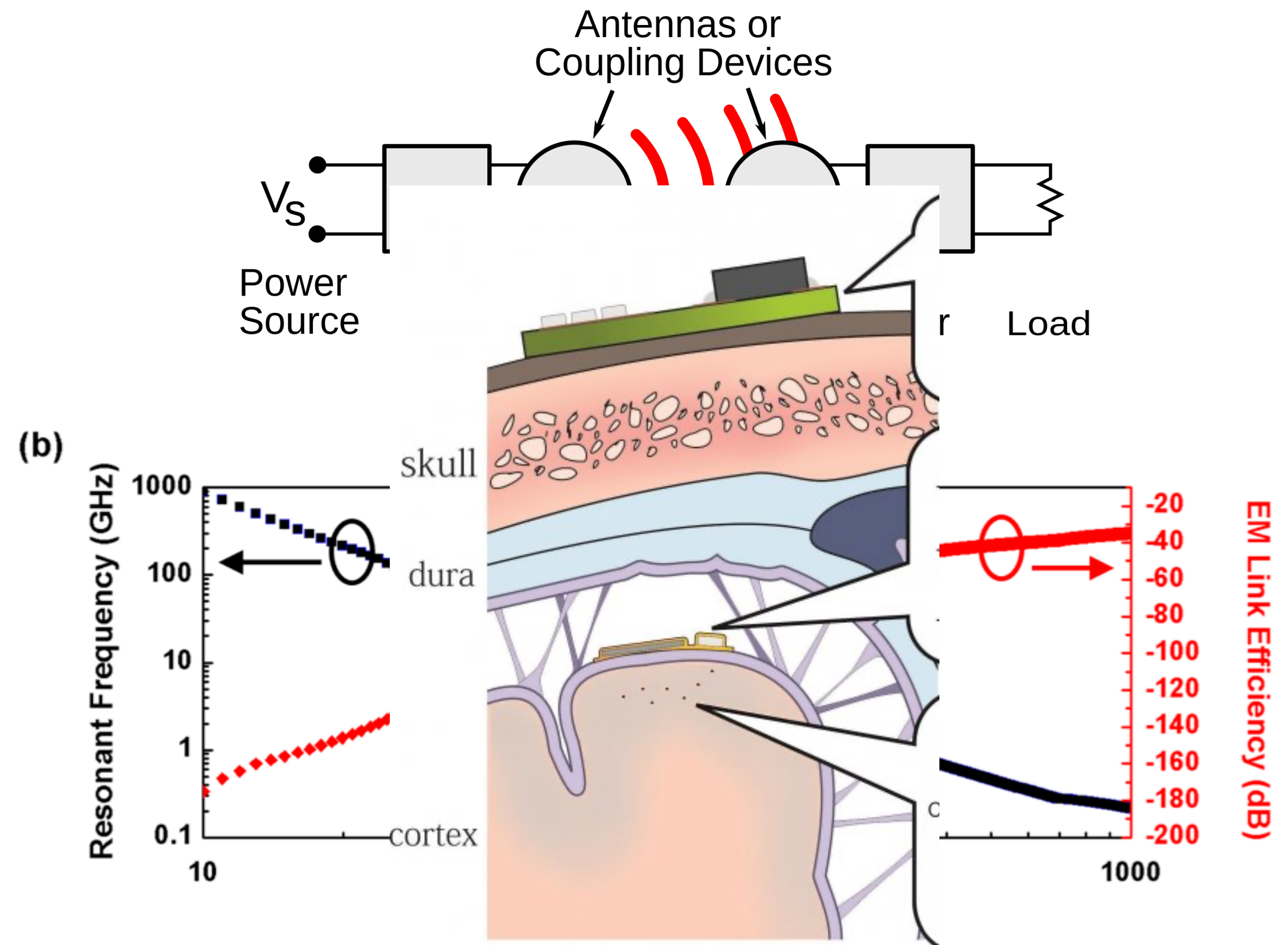
Addiction treatment had failed. Could brain surgery save him?

Experimental surgery has kept Gerod Buckhalter sober for more than 600 days. His success shows what may be possible.



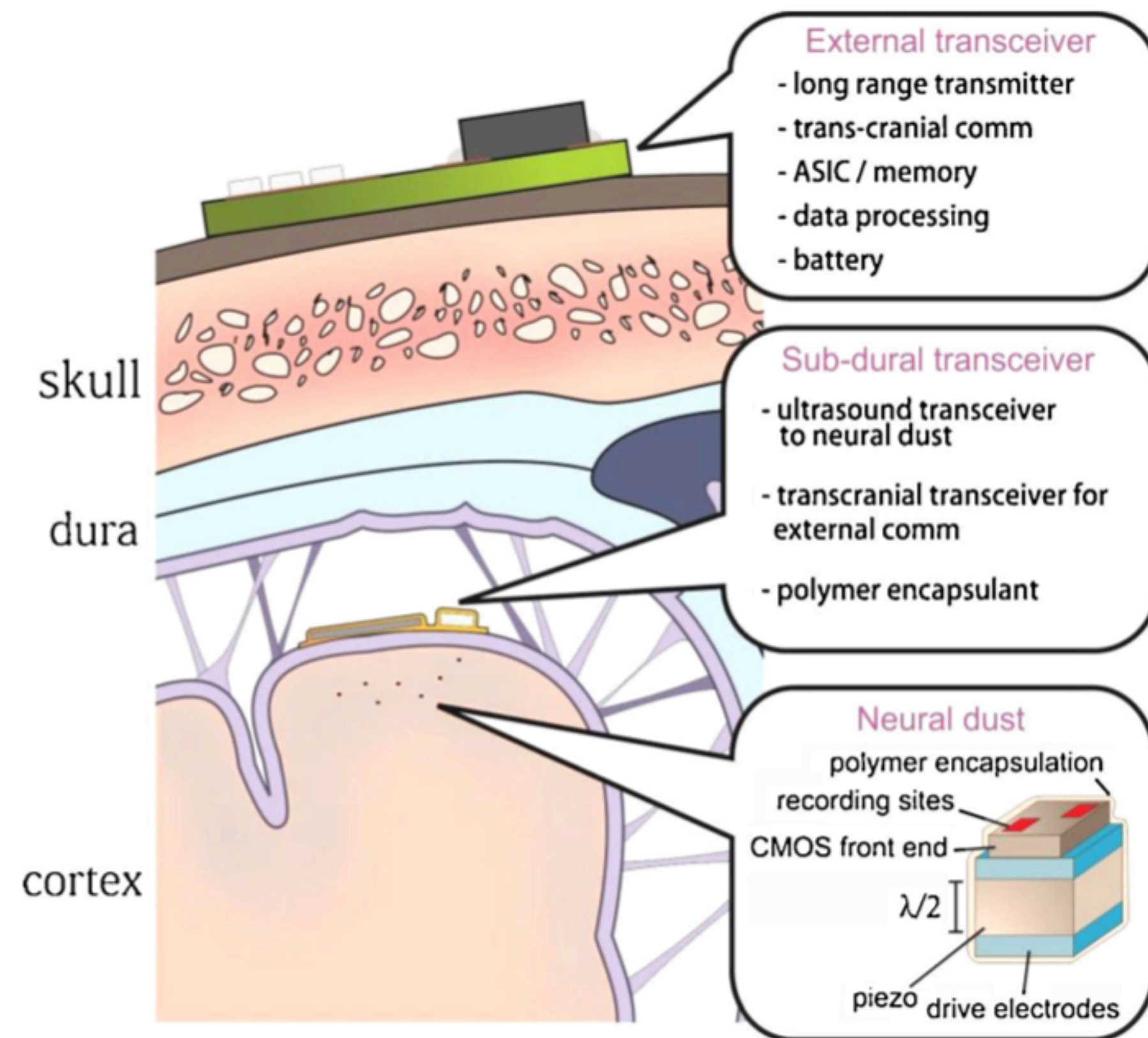
Implantable Devices

- **Size:**
 - Electrical potentials are measured differentially so small devices reduce the distance between recording electrodes ($V = E \cdot D$) → decreases measured voltage ($\therefore$ poor SNR)
 - Large devices cause tissue responses (e.g. glial scarring)
- **Power**
 - With electromagnetic radiation (e.g. RF) small nodes require high frequencies (a 100 μm antenna $\sim$ 10 GHz)
 - 10 GHz EM wave $\sim$ 3 mm wavelength $\gg$ 100 μm
 - Loss scales exponentially with frequency!



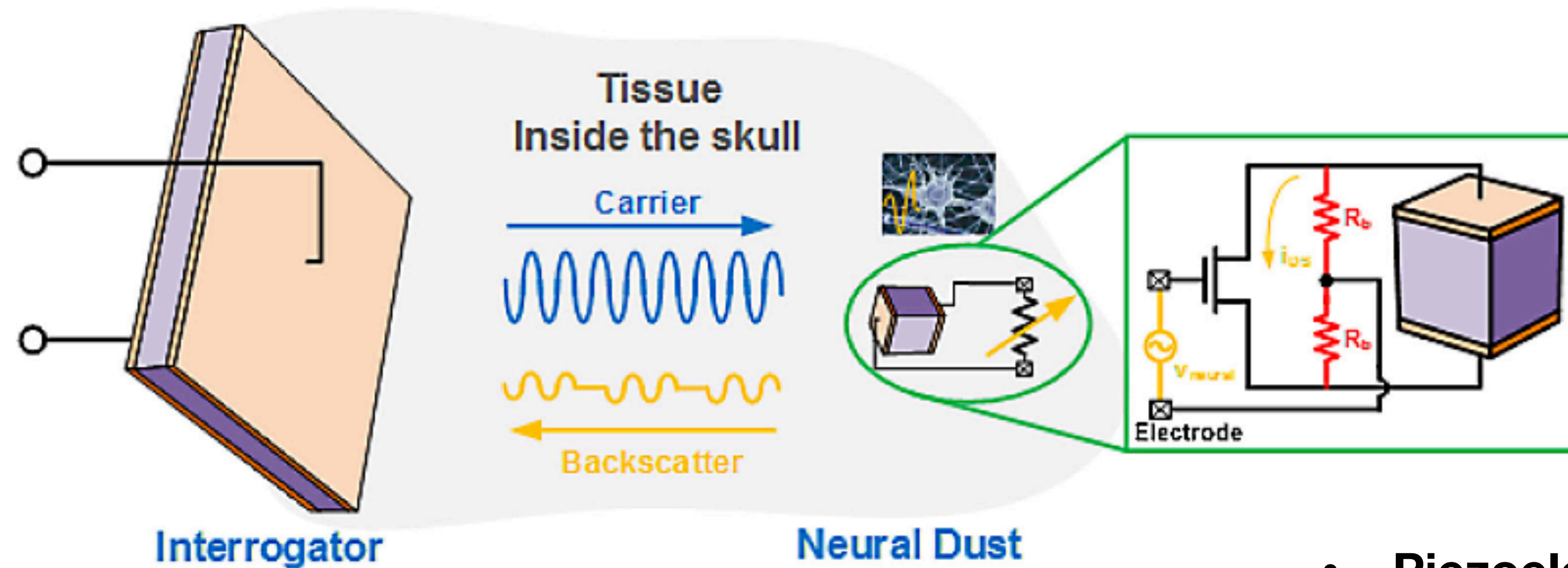
100 μm implant @ 2 mm in cortex = - 64 dB efficiency ($>10^6$)
 10 mW/cm² → **40 pW** of power (record¹ $\sim$ 10 μW / channel)

Solution: Acoustic Waves!



- **Ultrasound for power and communication**
 - Much lower loss (0.5 dB/cm/MHz)
 - 10 MHz = 1 dB loss (20%) (c.f. EM = 20 dB ~99%)
 - Higher allowed intensity (FDA) ~ 100 mW/ cm² ~ 10x EM
 - Speed of sound is slower (~1500 m/s) $\therefore$ wavelength is shorter!
 - 10 MHz = 150 μ m

Principle of Operation

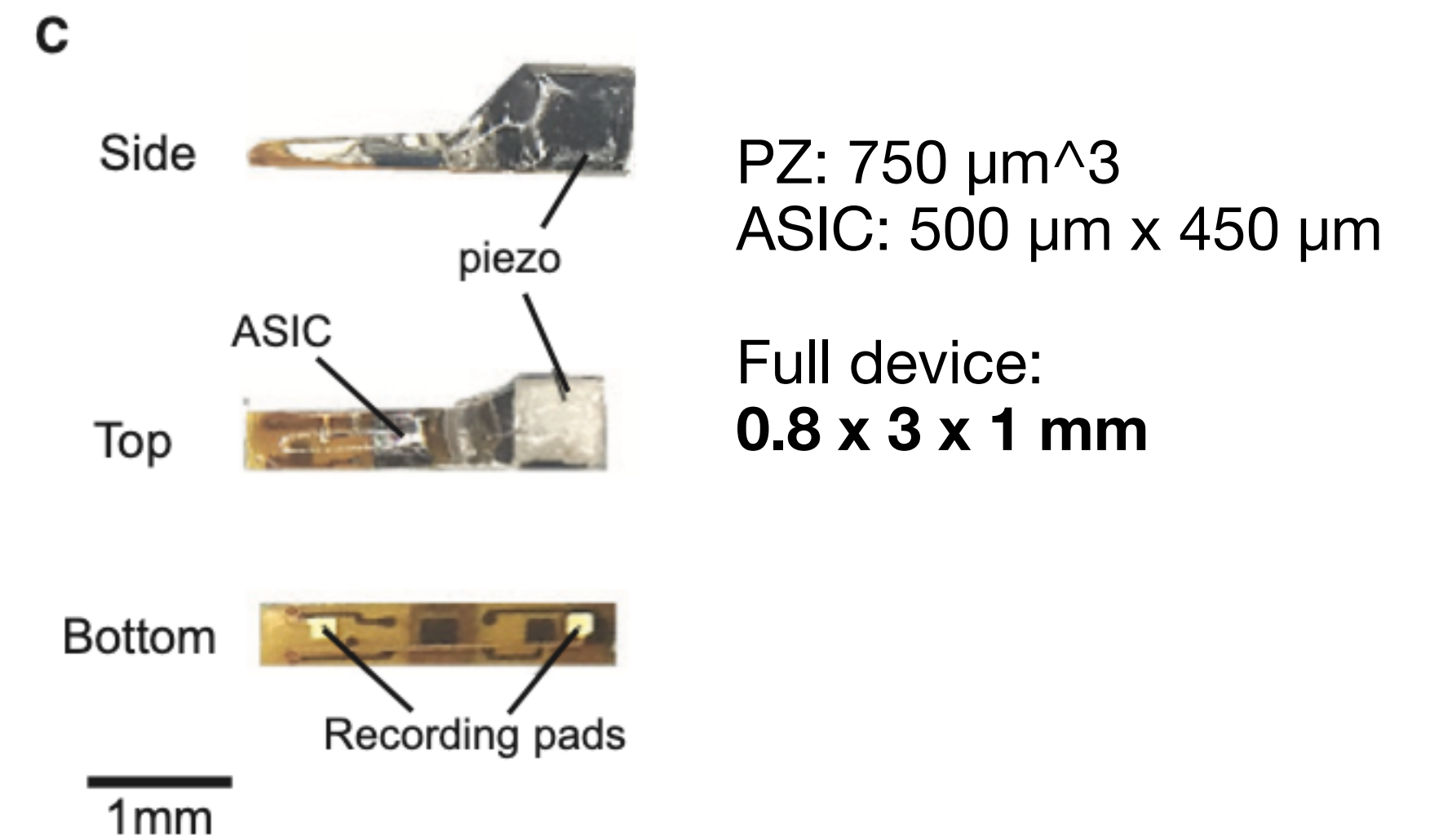
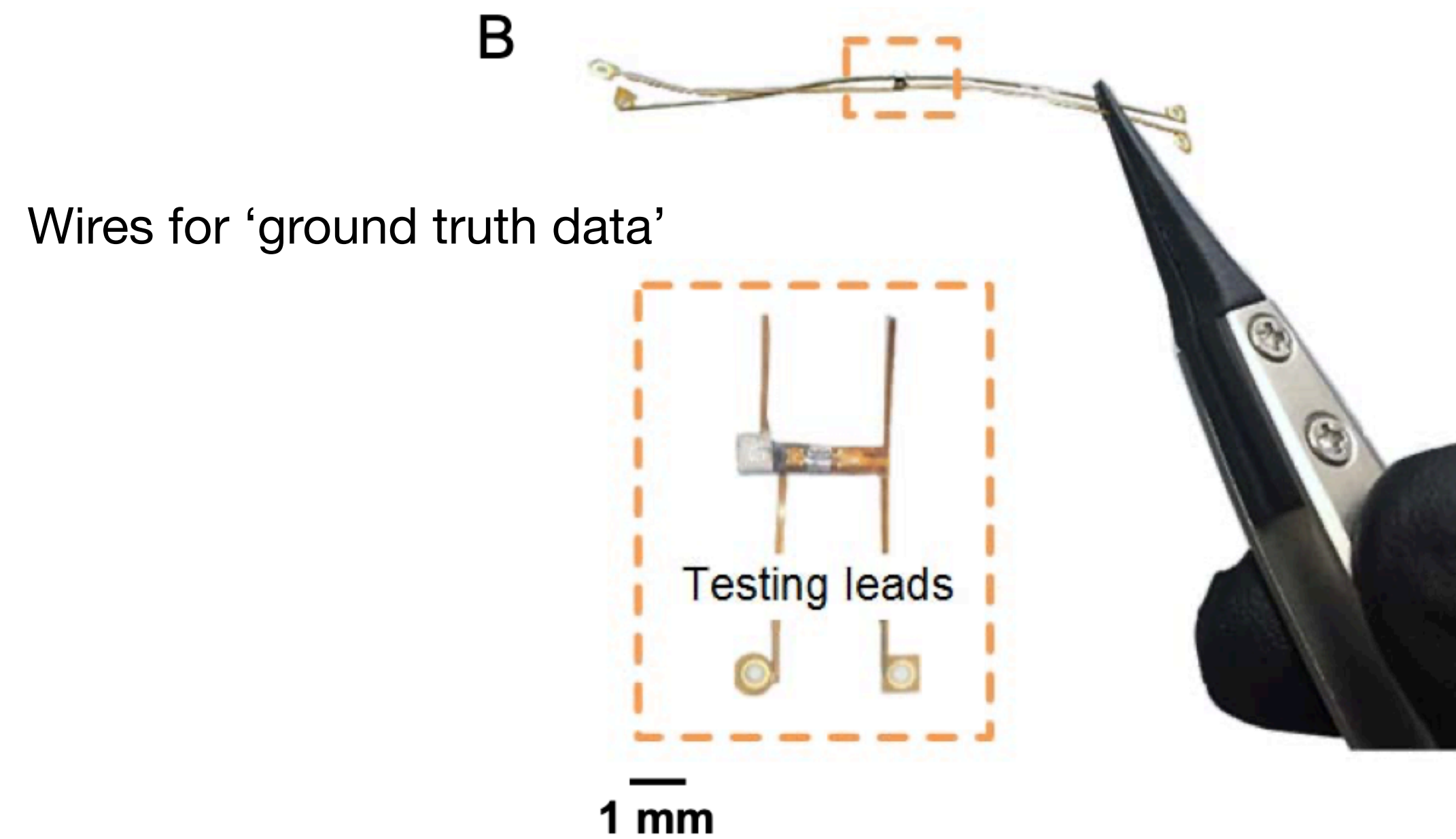
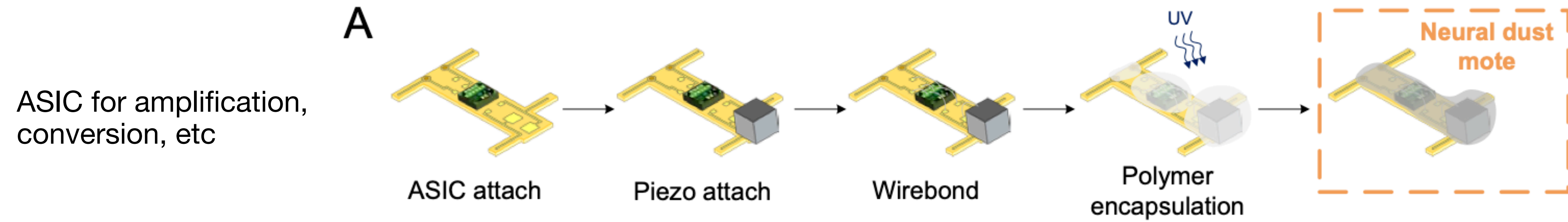


For a 100 μm dust node 2mm into the brain $\sim 500 \mu\text{W}$ ($10^7 > \text{EM}$)

- **Piezoelectrics**

- A material that couples mechanical properties and electrical properties
- e.g. squeeze you can generate a voltage or apply voltage and you can squeeze
- PZT (lead zirconate titanate 🤖) or BaTiO
- A circuit that can modulate the impedance of the PZ dependent on an action potential

Neural dust assembly

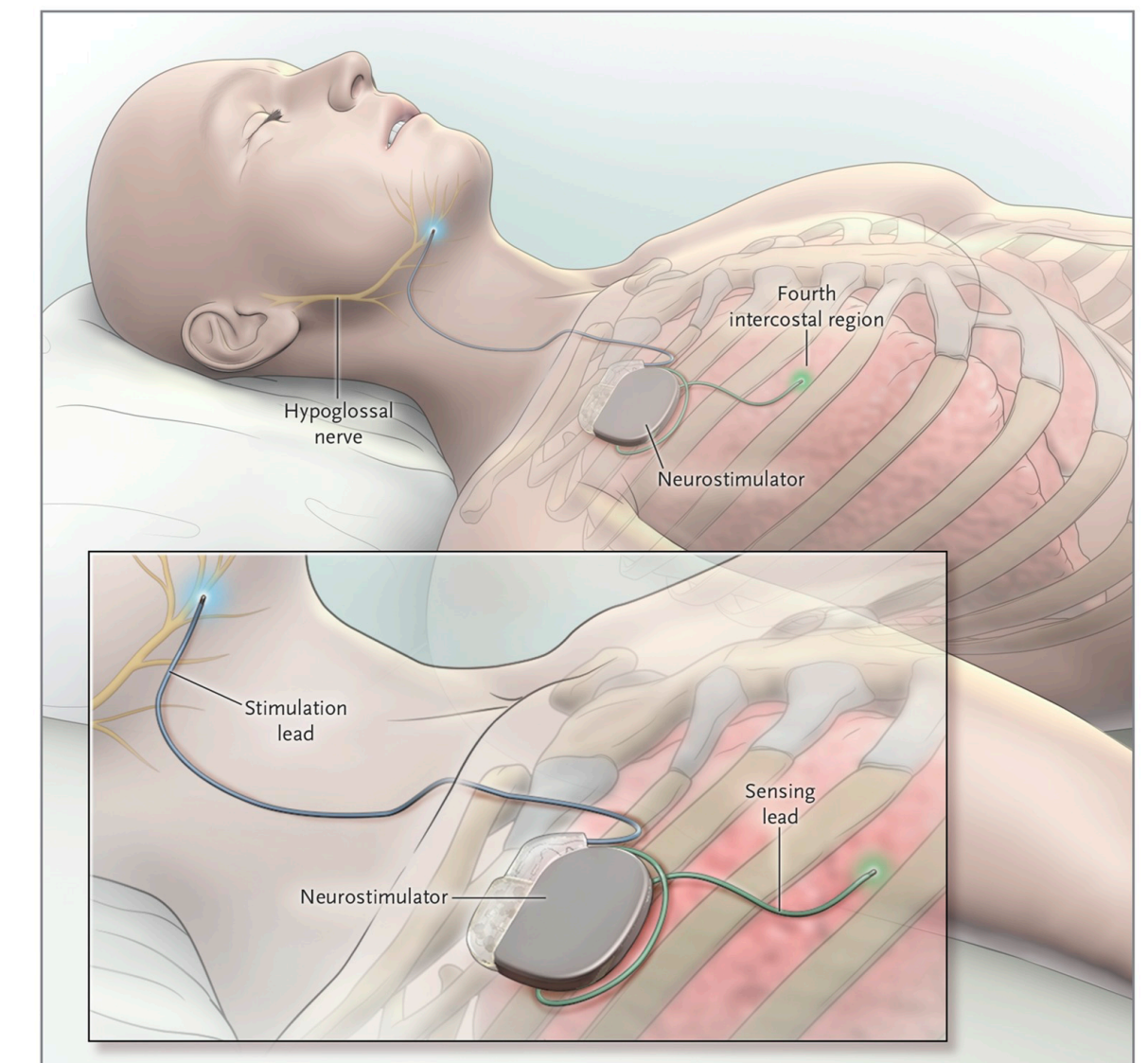
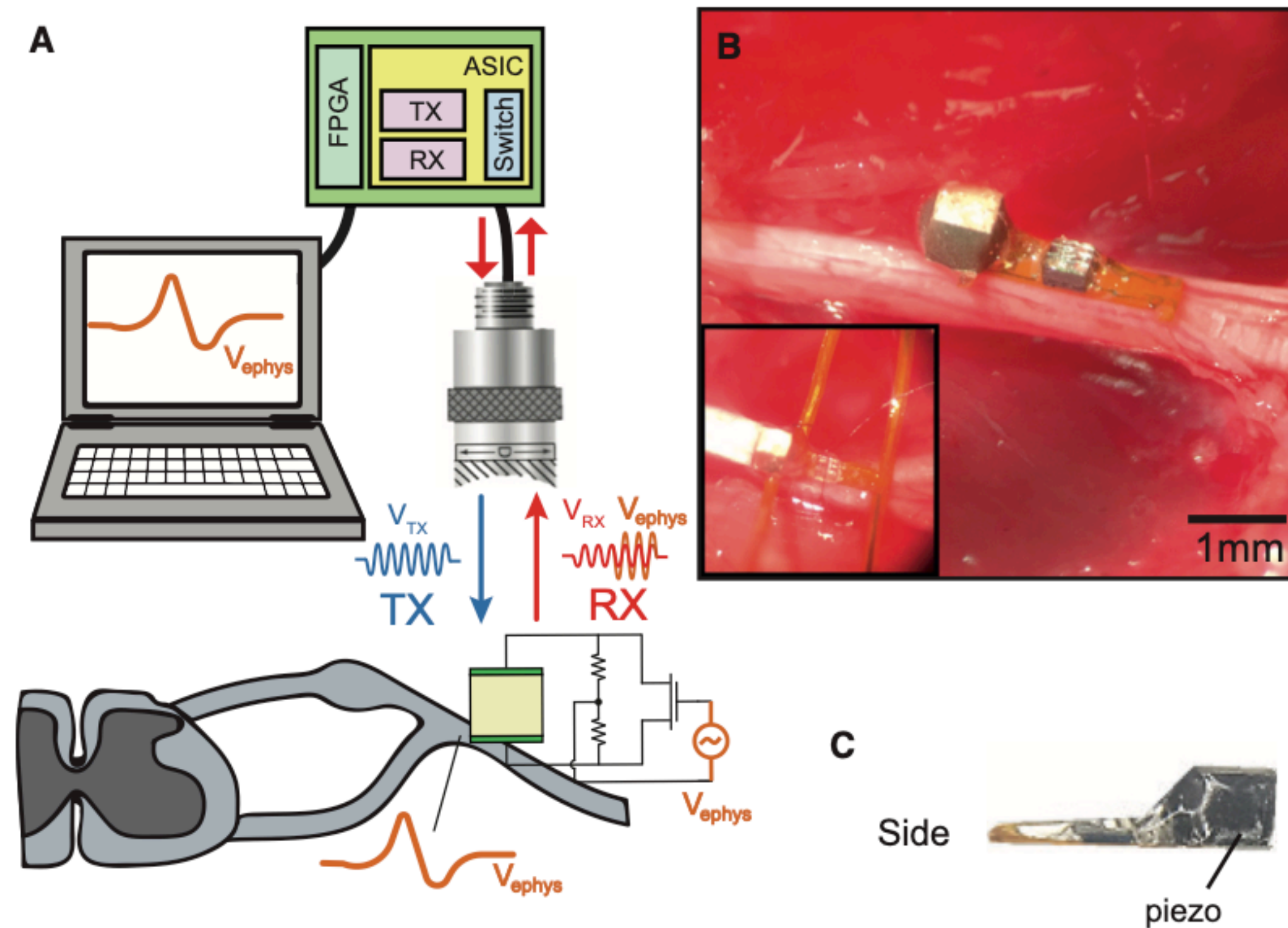


Peripheral Nervous System

Applications:

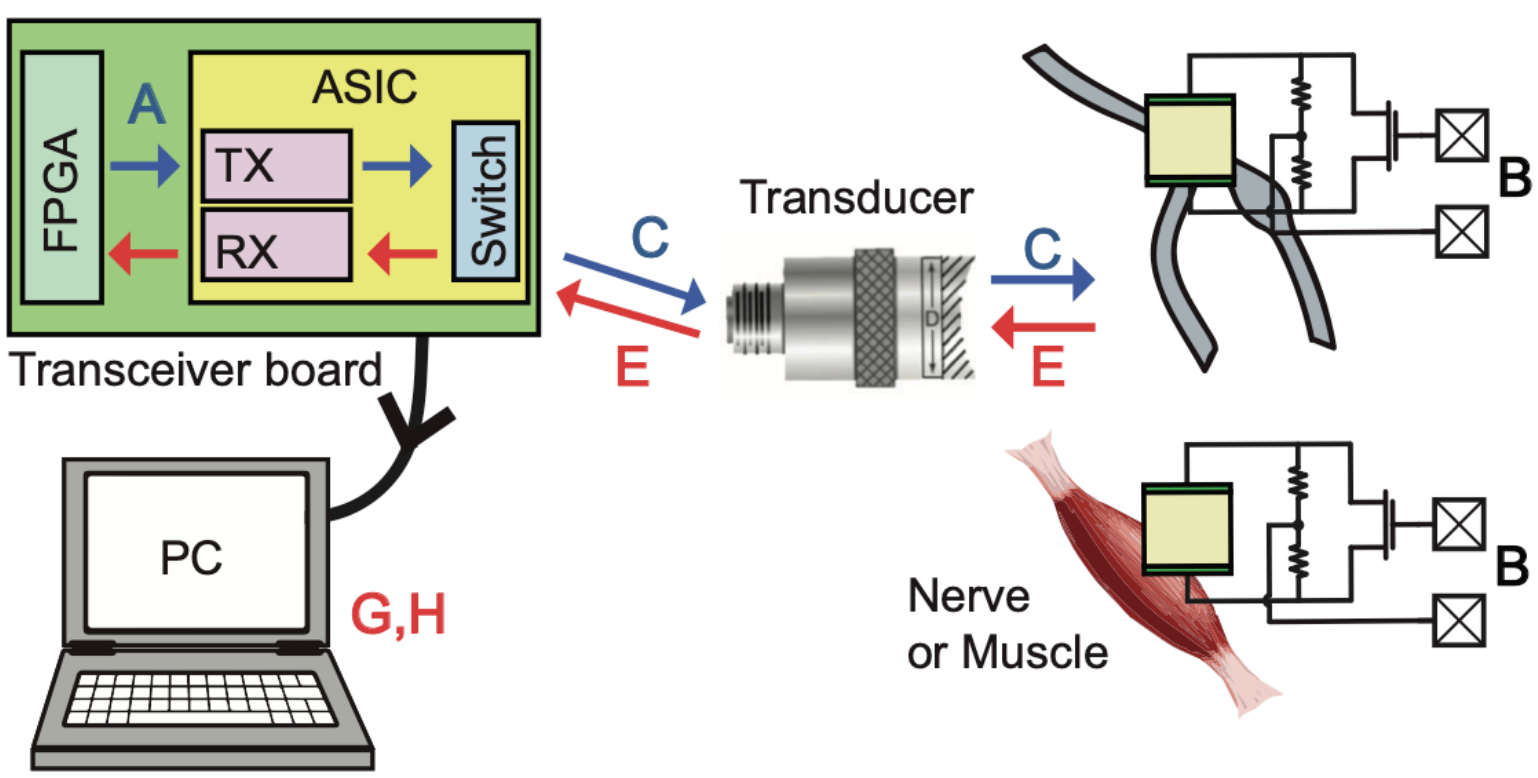
- Sleep apnea
- Control of bladder function in paraplegic patients
- Diabetes
- Rheumatoid Arthritis

Famm et al., *Nature* (2013)

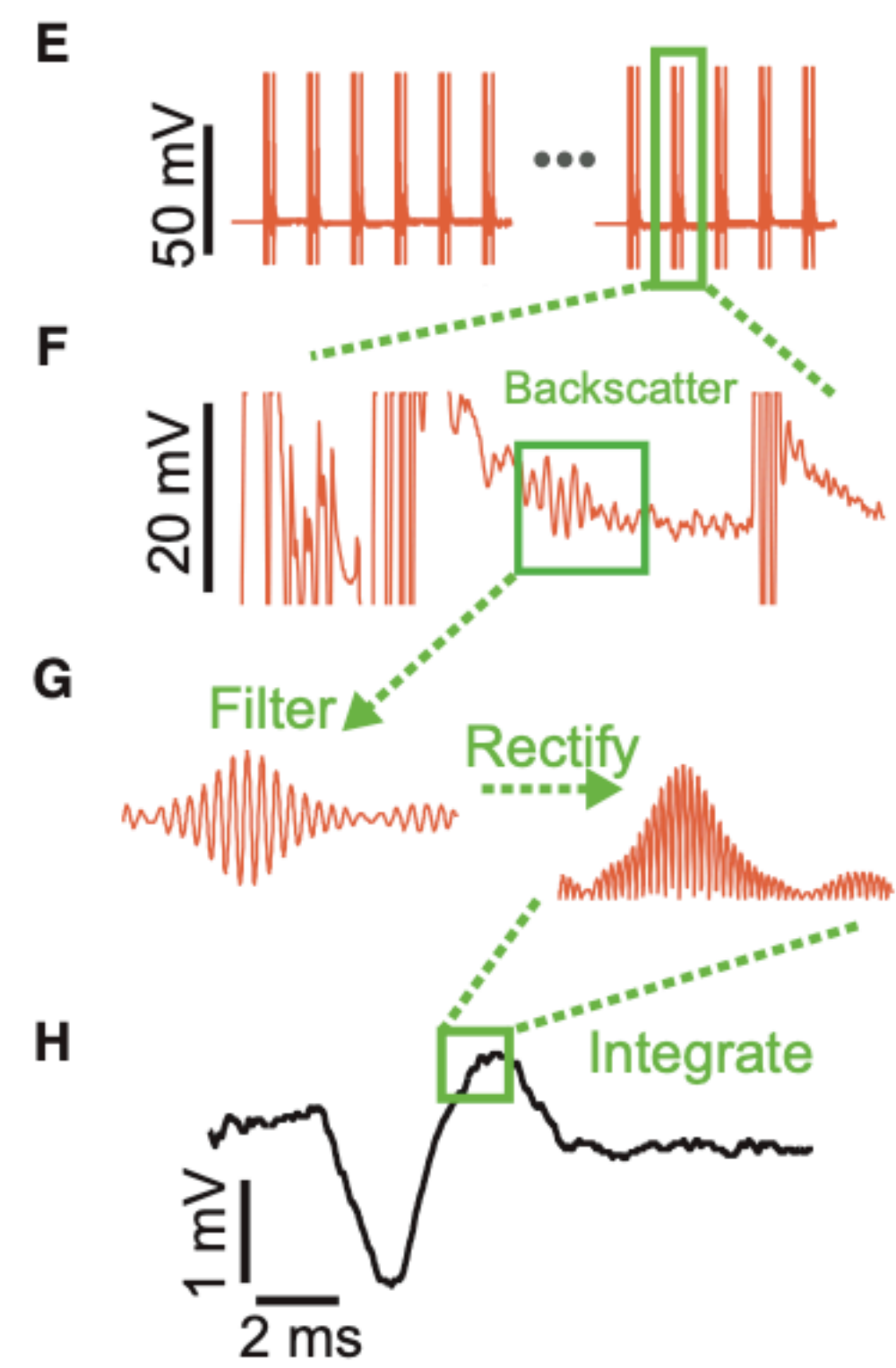
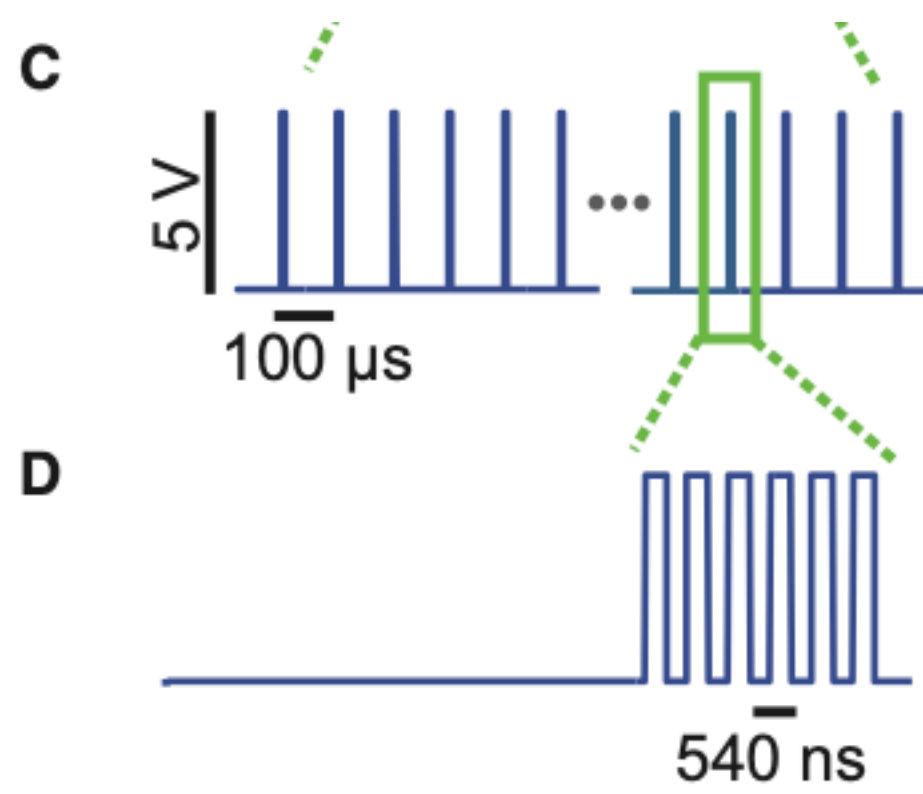


Seo et al., *Neuron* (2016)

Communication

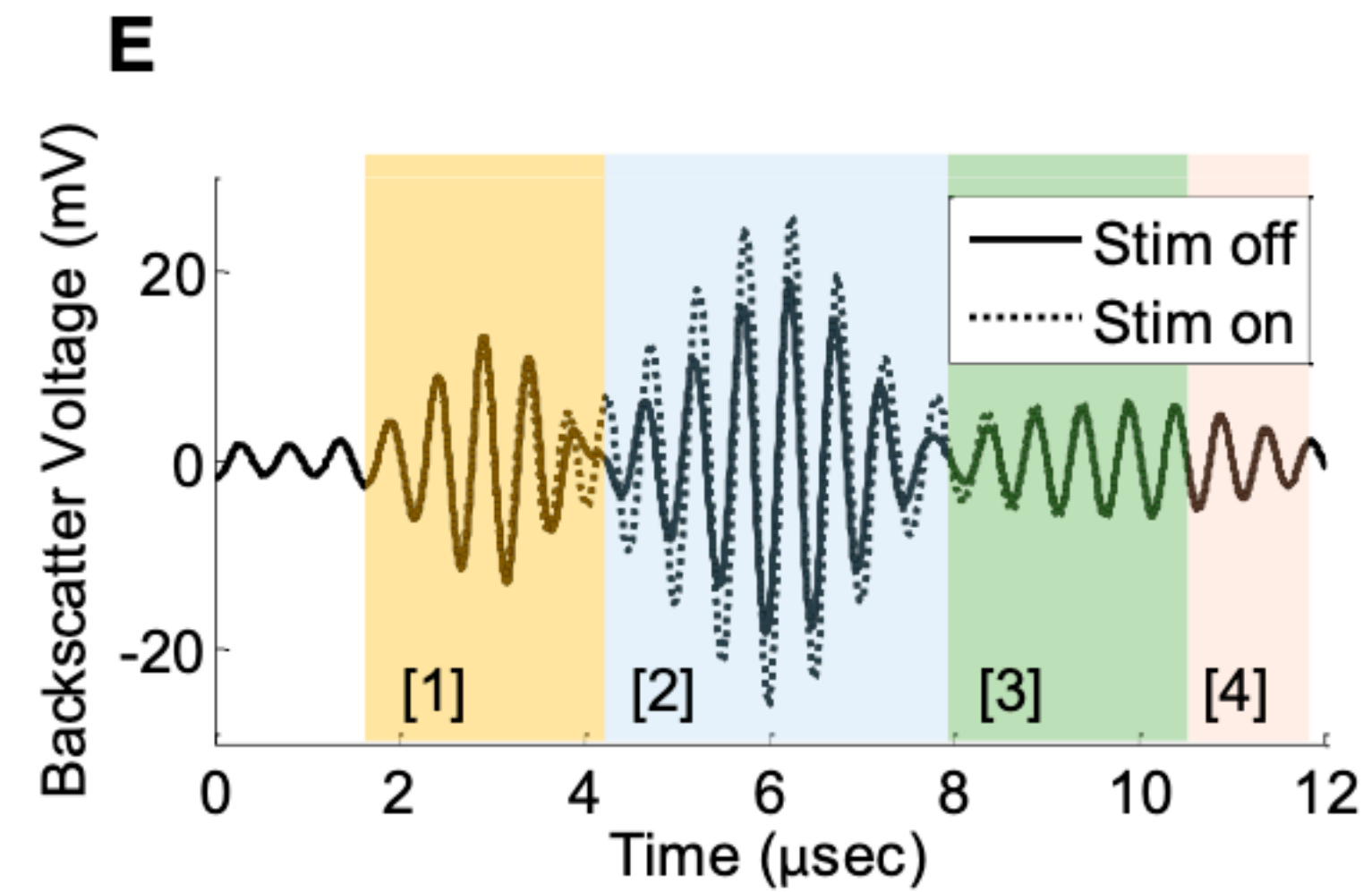


Send in pulses every 6 ns then listen!

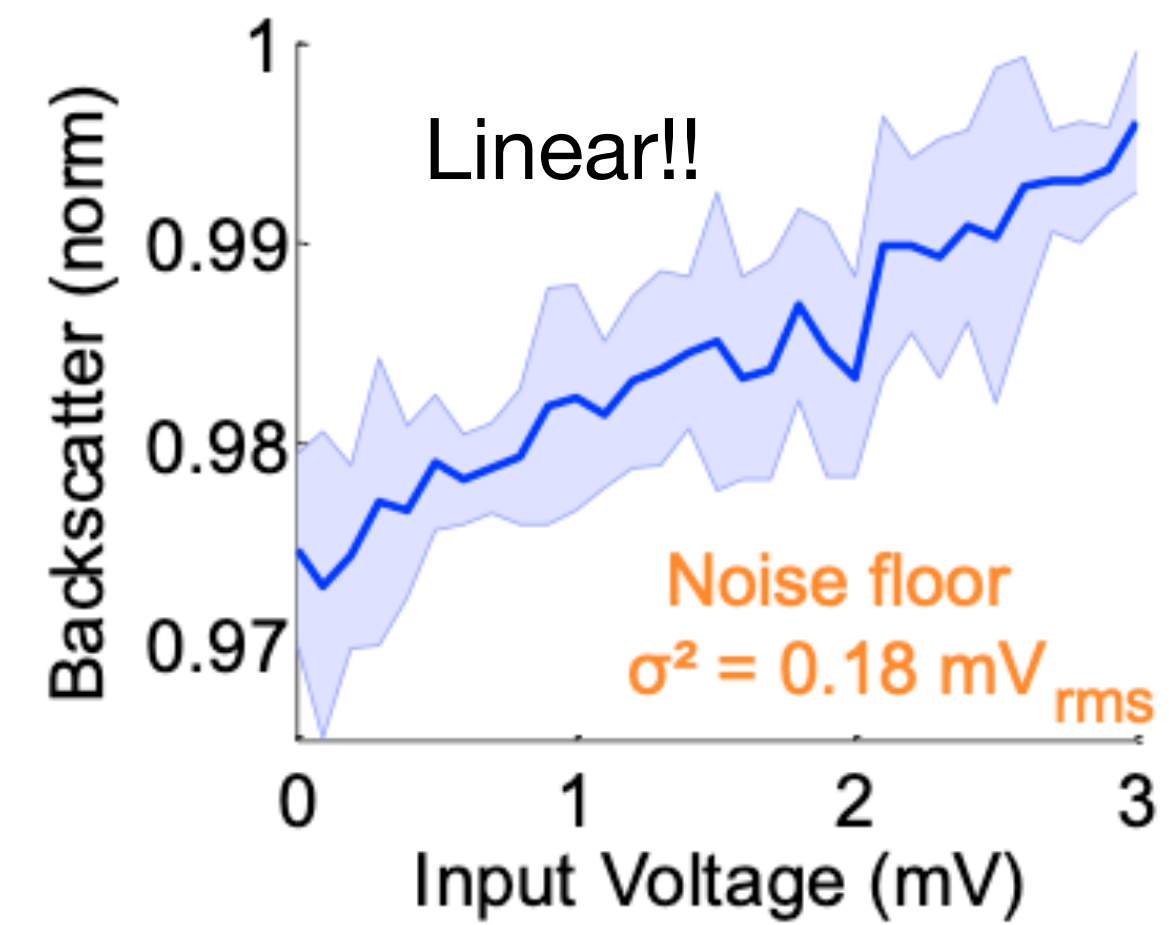


10 KHz readout!

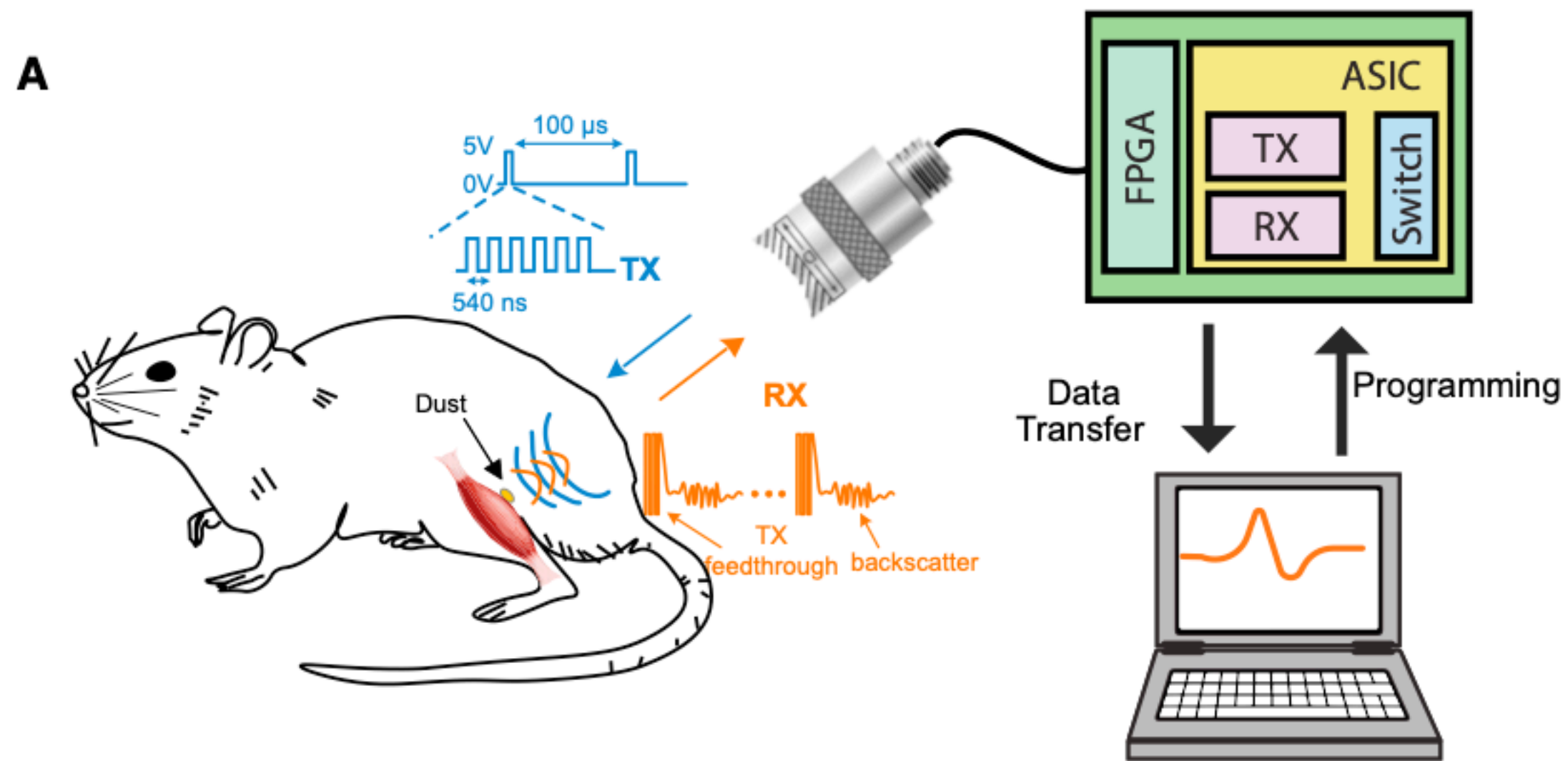
Calibration



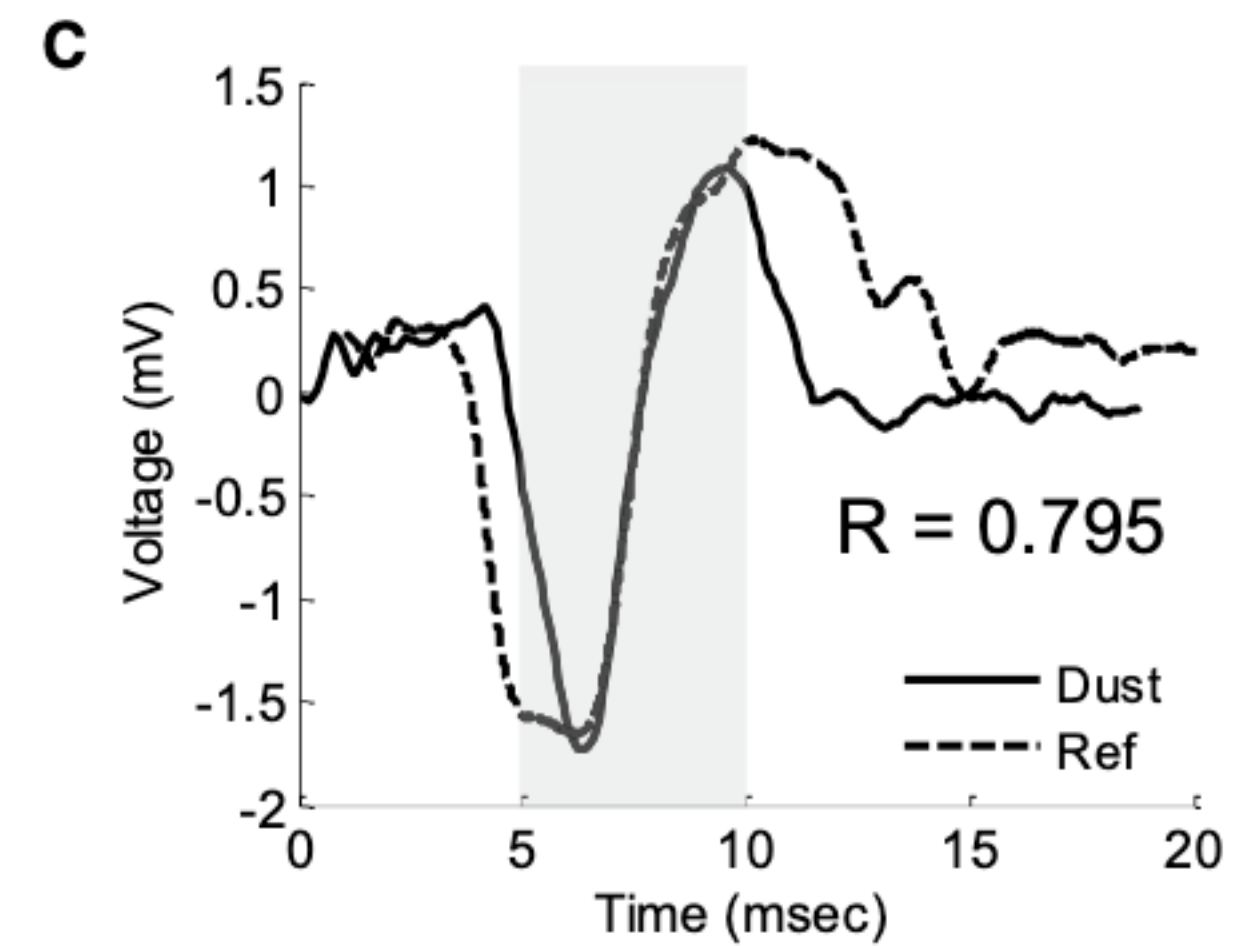
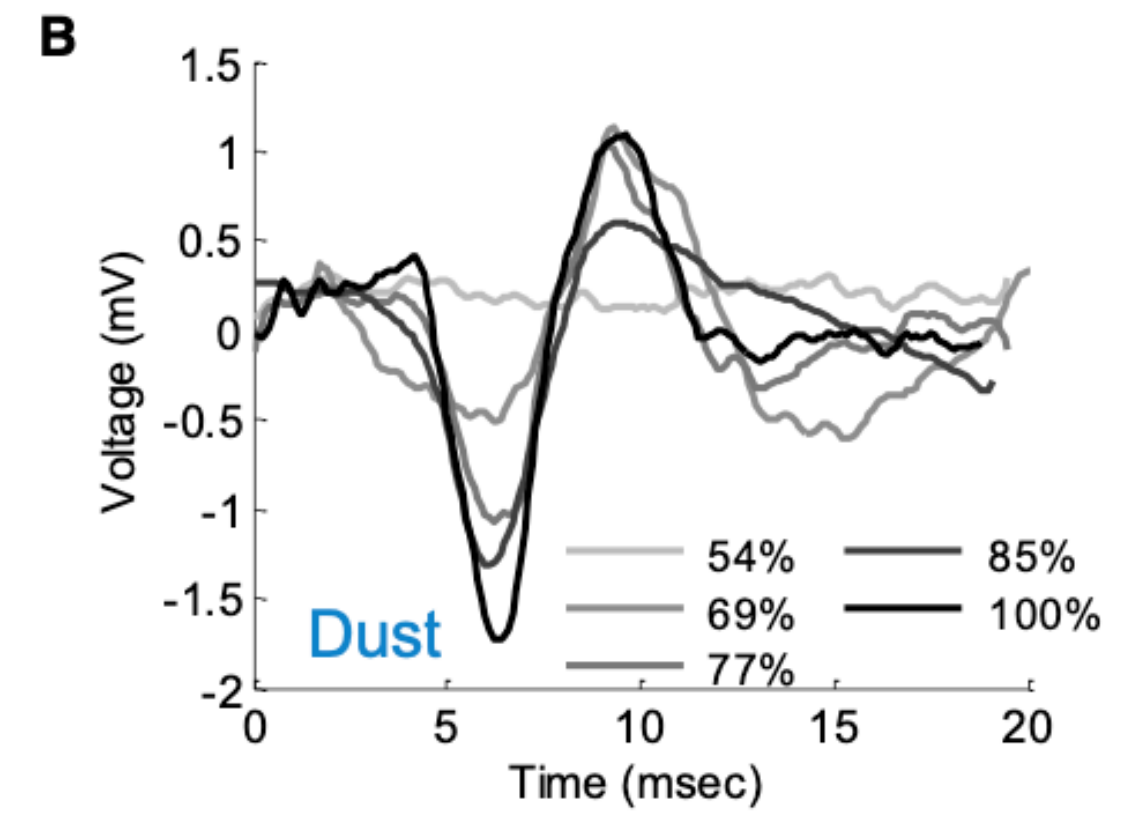
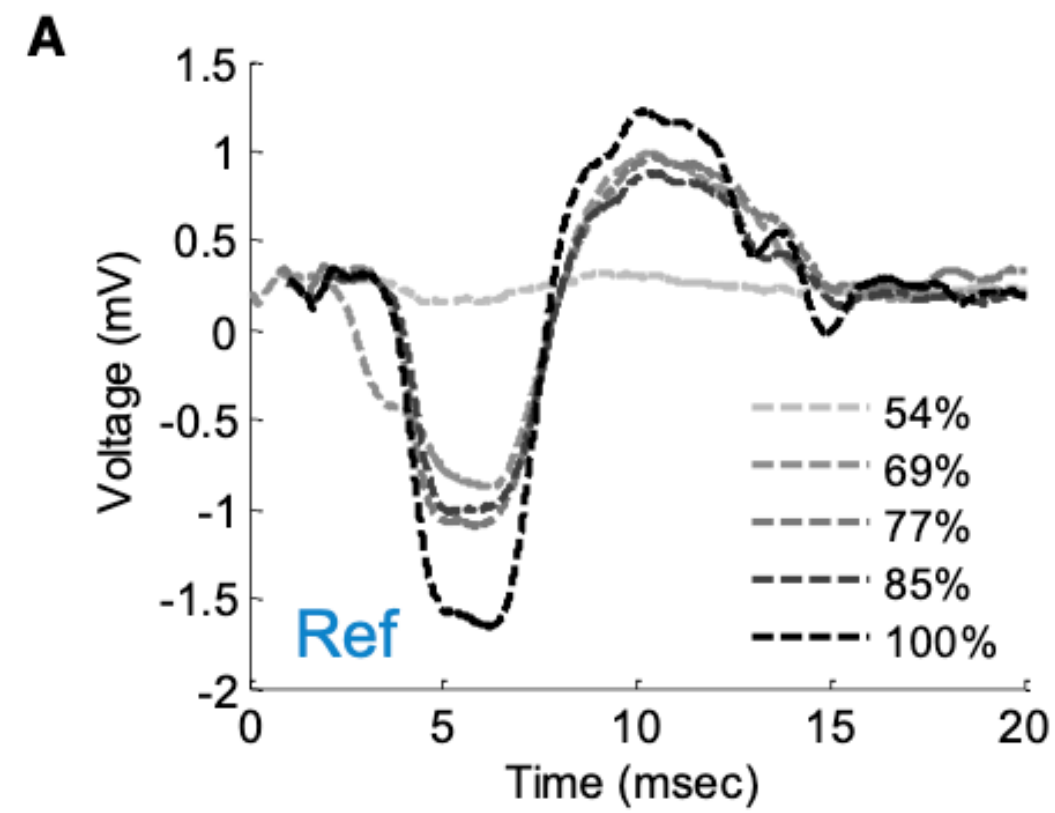
1. Water-Polymer boundary
2. Top of PZ crystal
3. Piezo-PCB boundary
4. Back of PCB



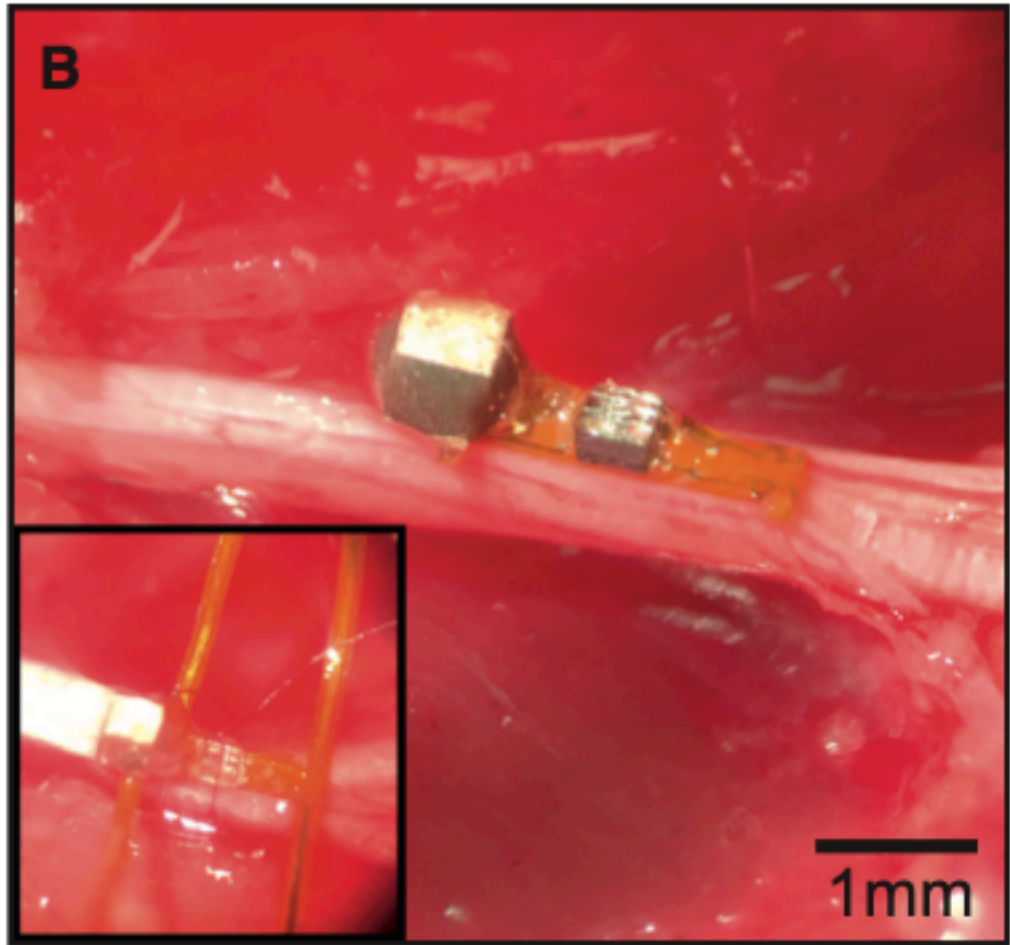
Electromyograms



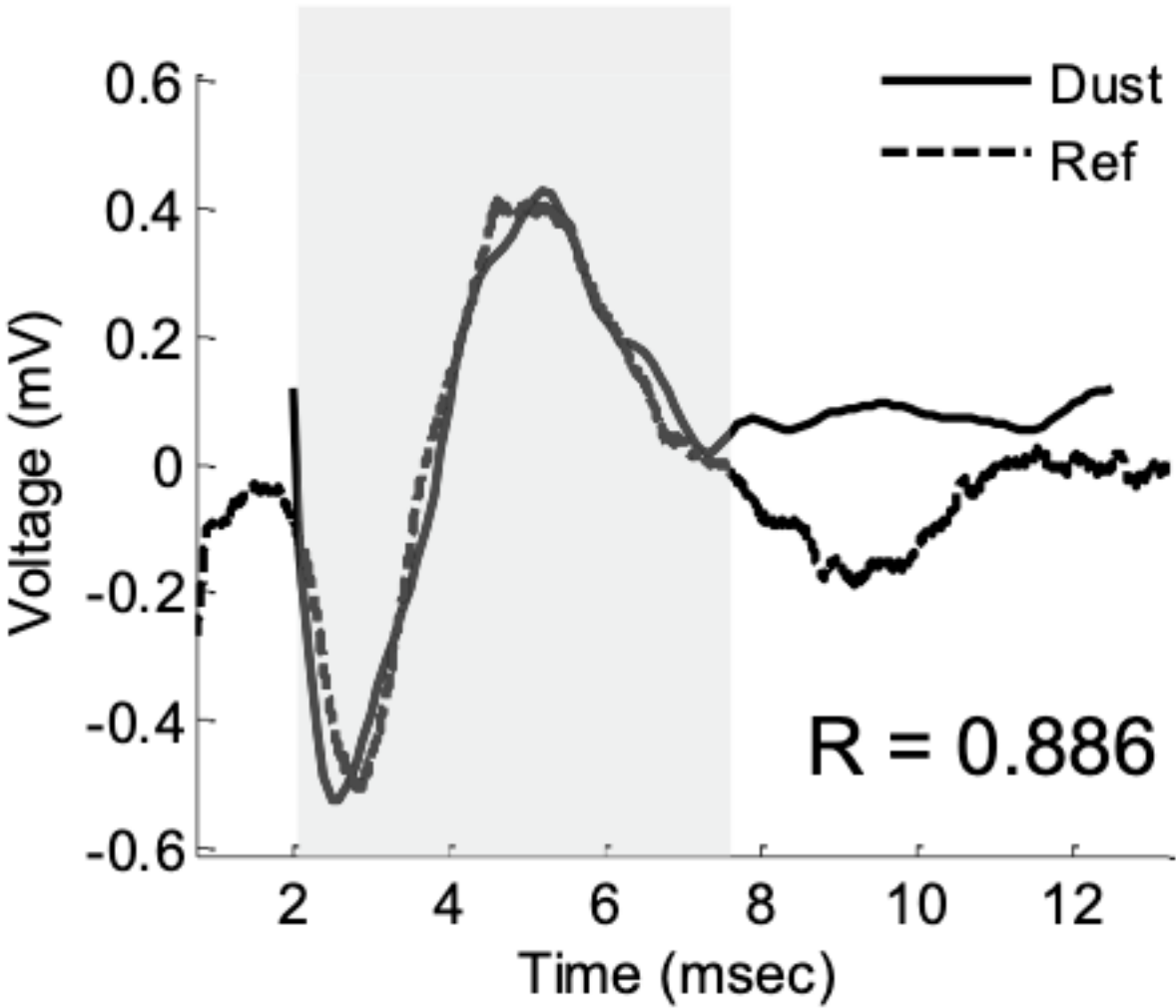
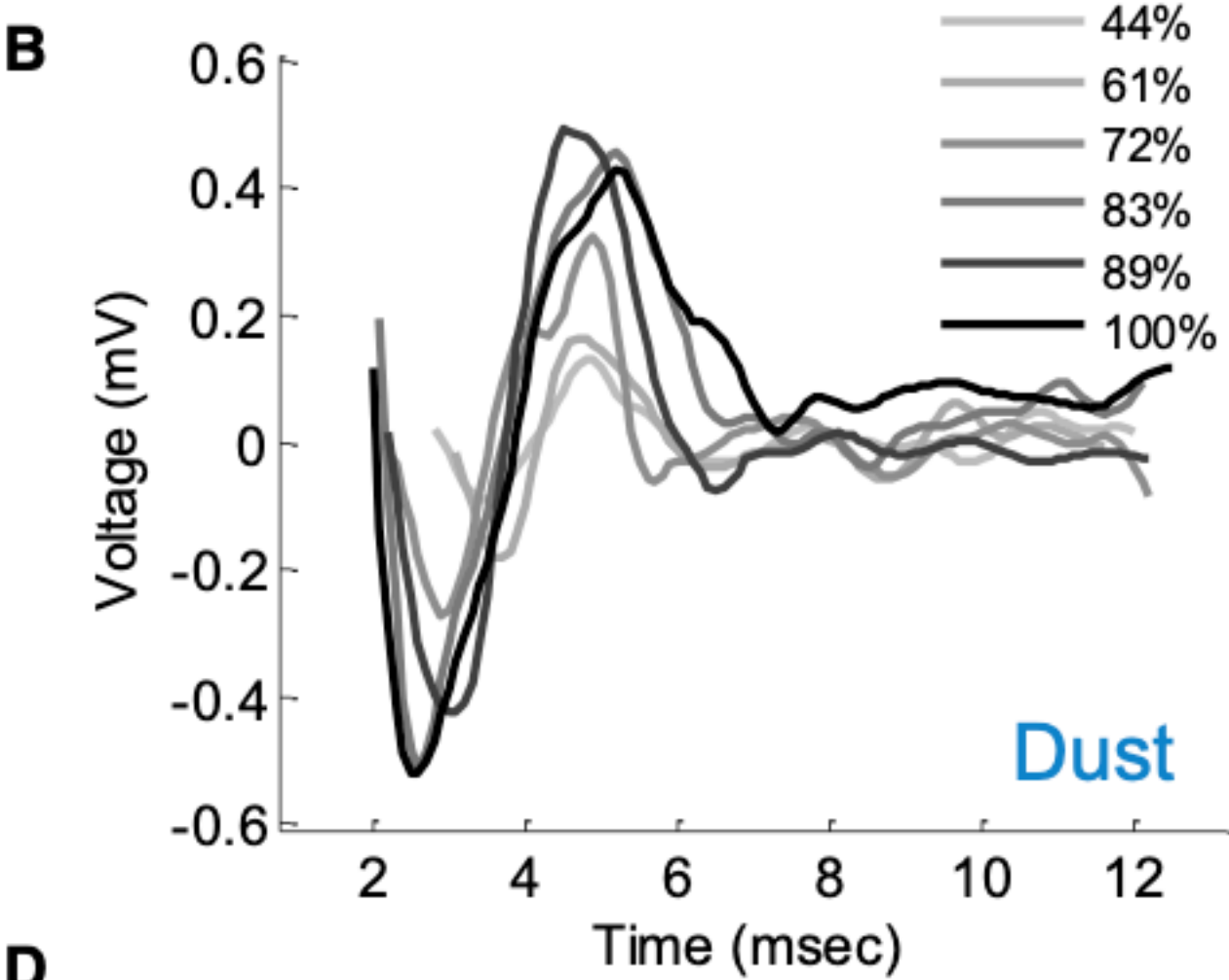
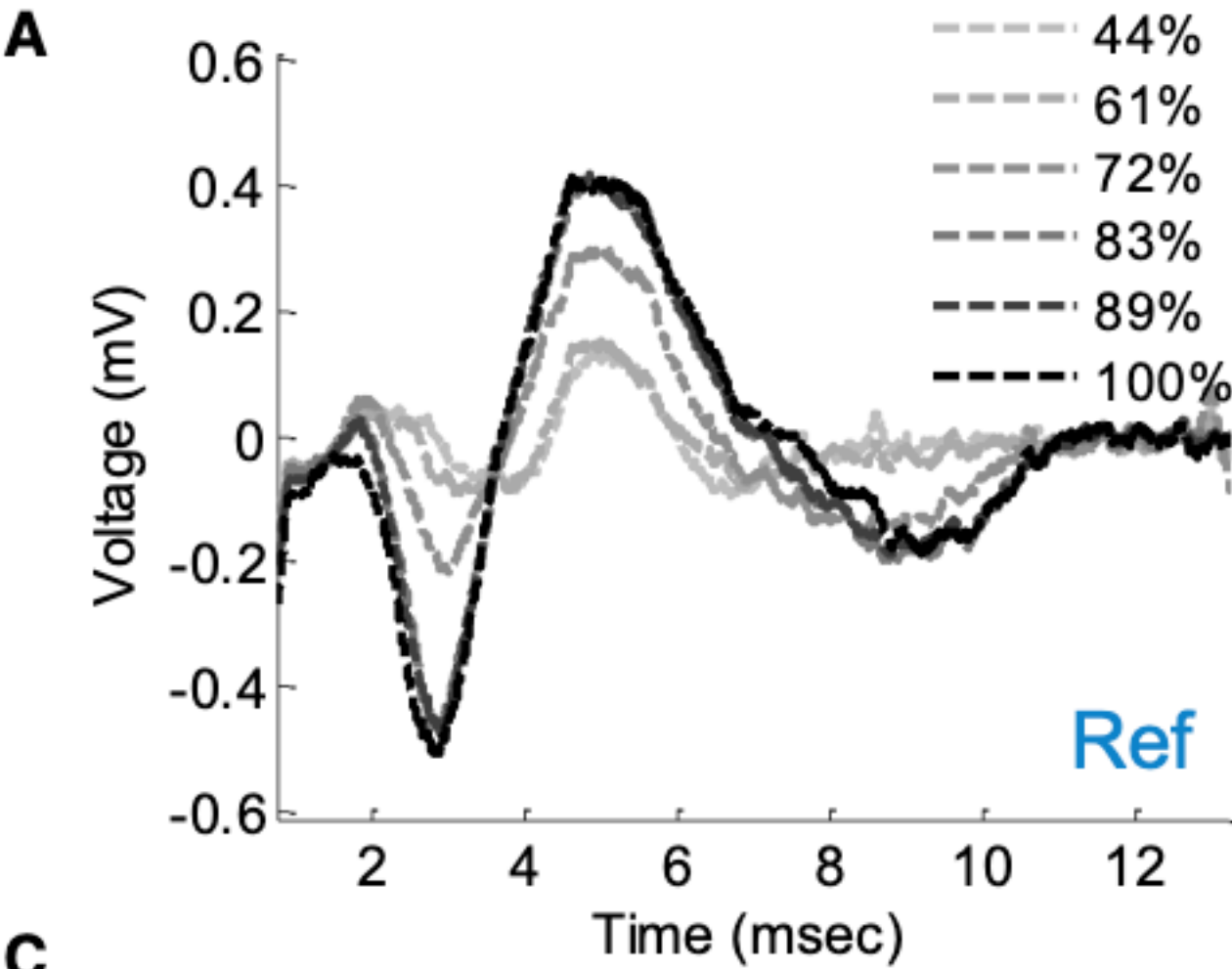
- ND on gastrocnemius muscle of rat under anesthesia
- Stimulate the muscle to invoke a response



Electroneurograms

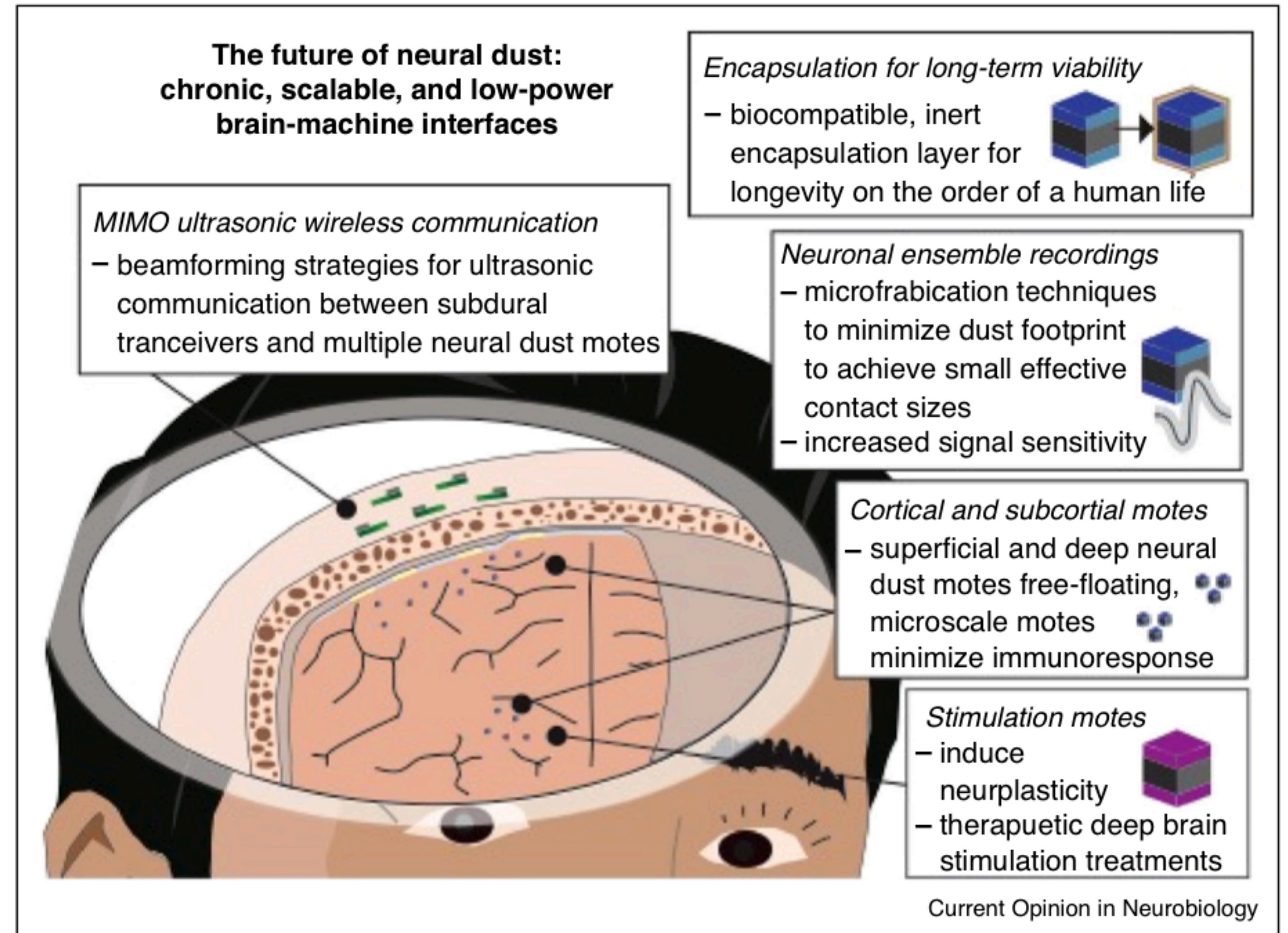


Neural Dust on sciatic nerve



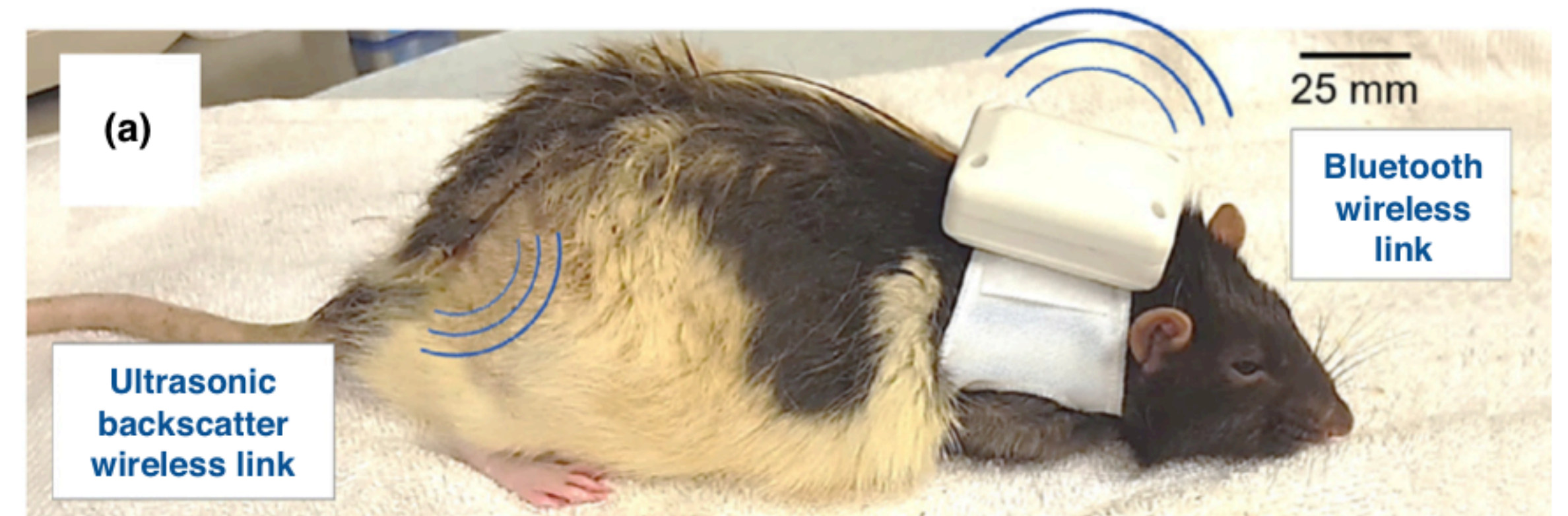
Technical Challenges

- Reducing size down is hard!!
 - Smaller piezoelectrics → worse SNR → more stringent CMOS
- Biocompatibility
 - Move away from PZT (but lower piezoelectric effect)
 - Polymers permeable to water vapor ∴ new encapsulation materials (SiC)
- Packaging
- Alignment and online motion correction

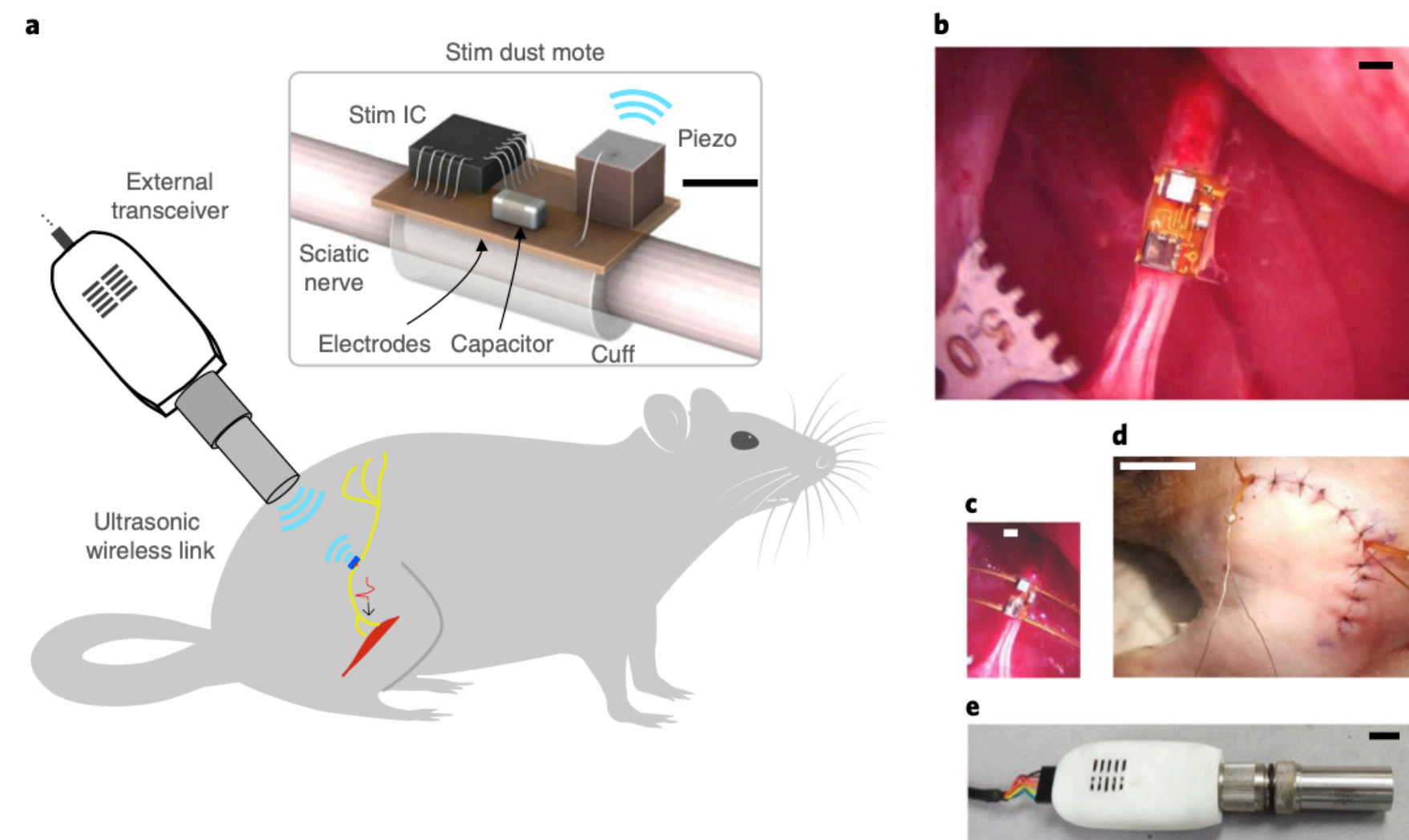


Where can this go?

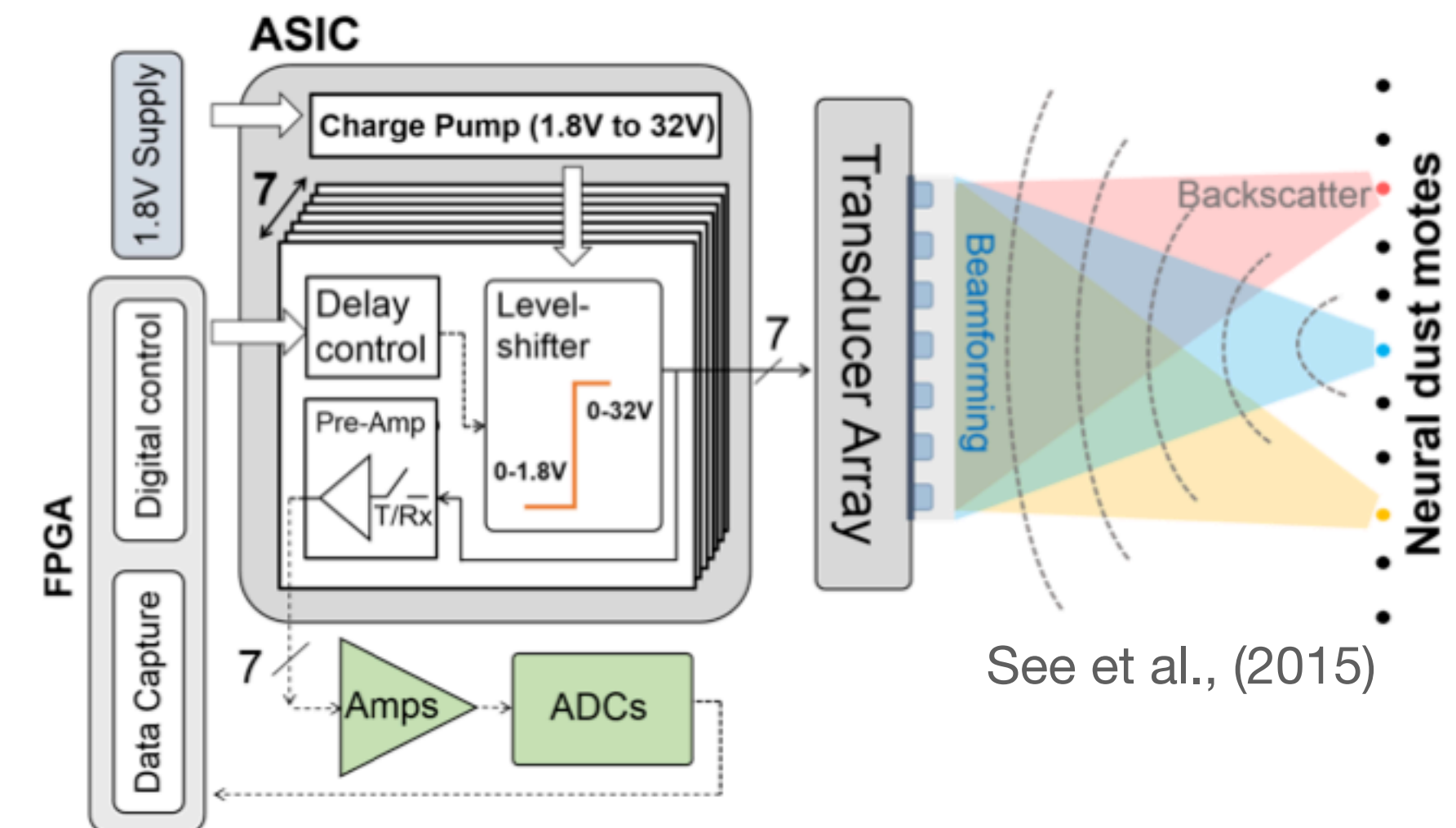
- Wearable ultrasound transducer with bluetooth link
- Transducer arrays for beam steering to interrogate multiple neural dusts motes + correct misalignment
- StimDust!



Piech et al., IEEE (2017)



Piech et al., Nat Bio Eng (2020)



Oxygen Sensor!

nature
biotechnology

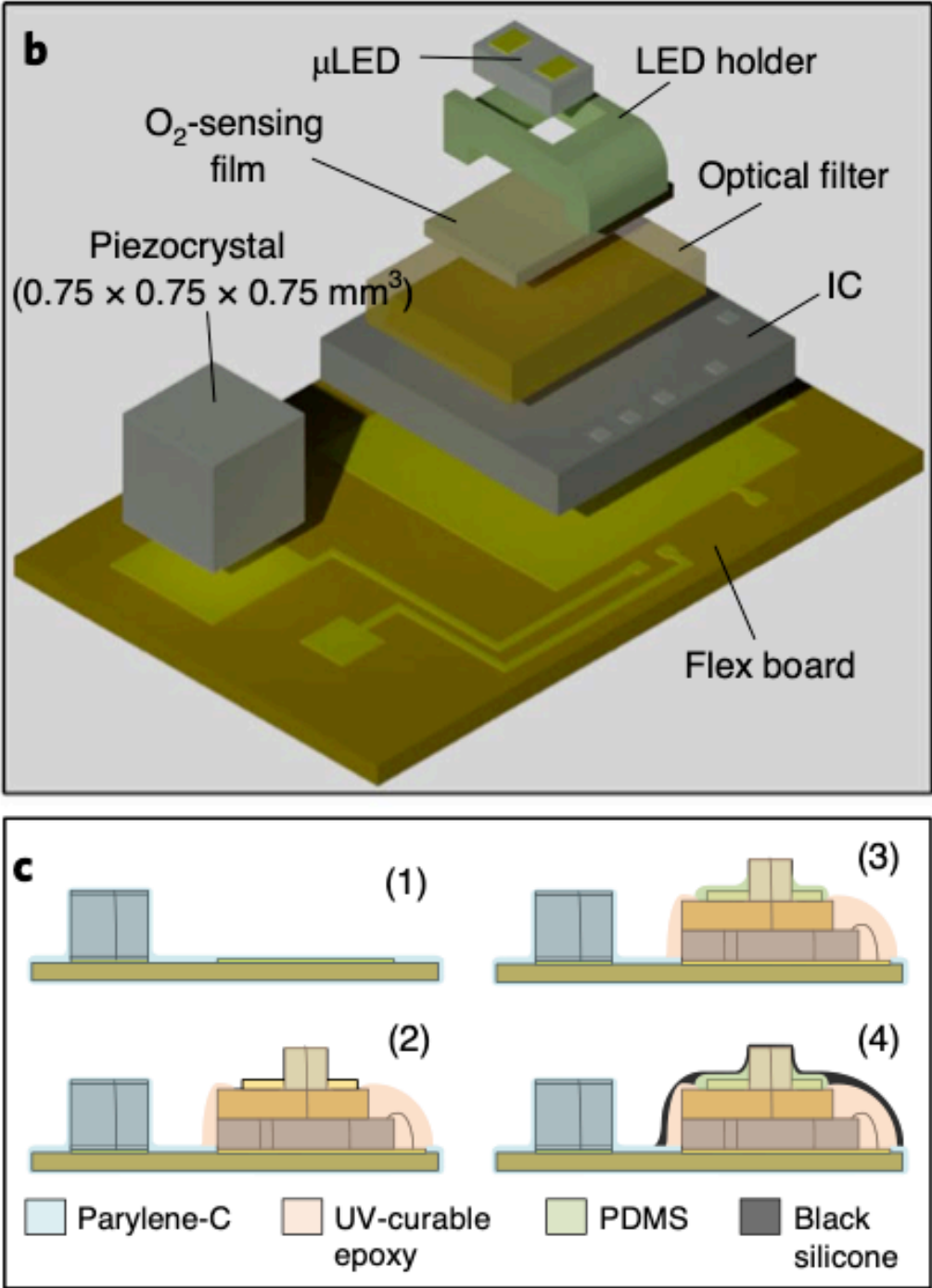
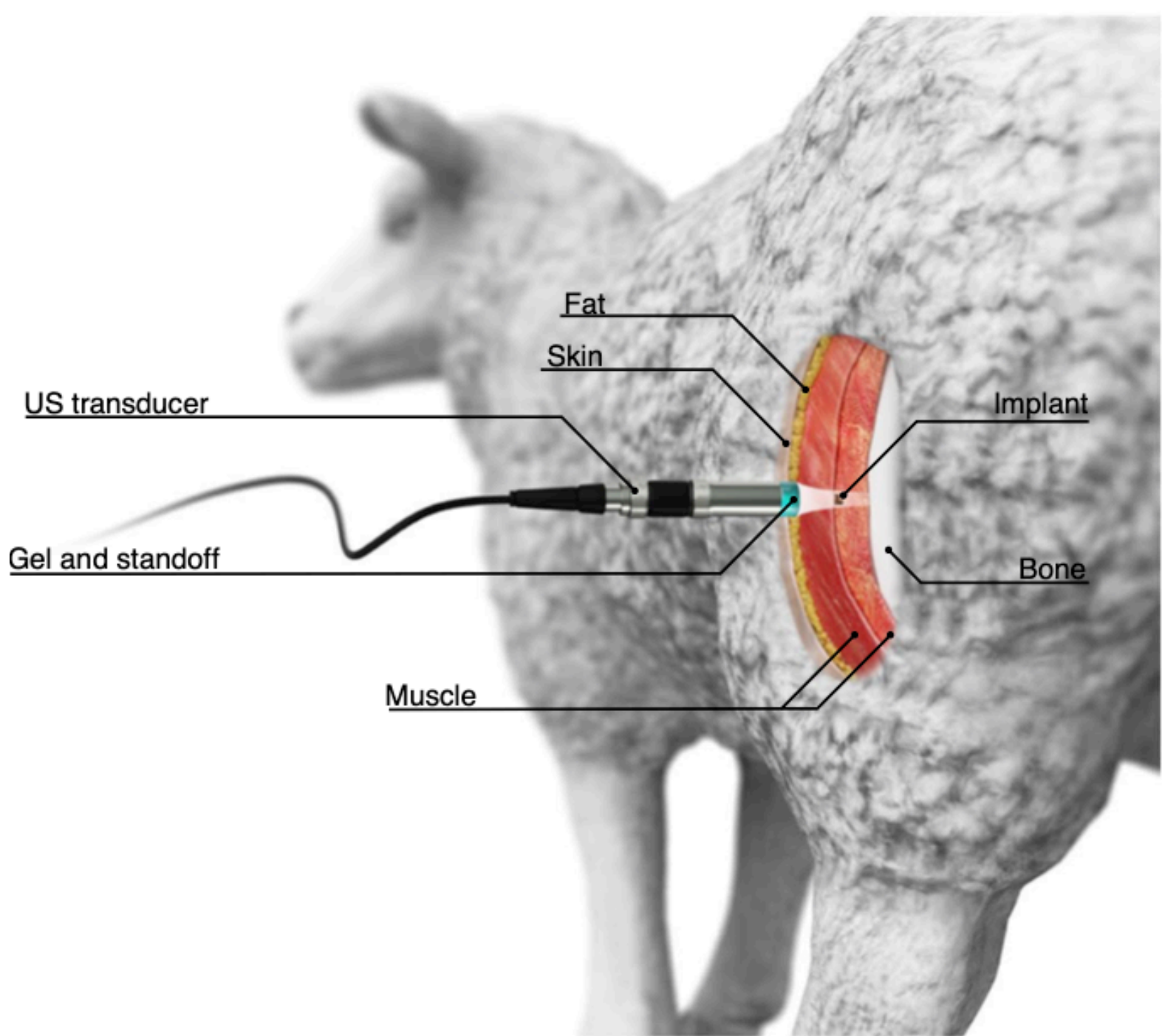
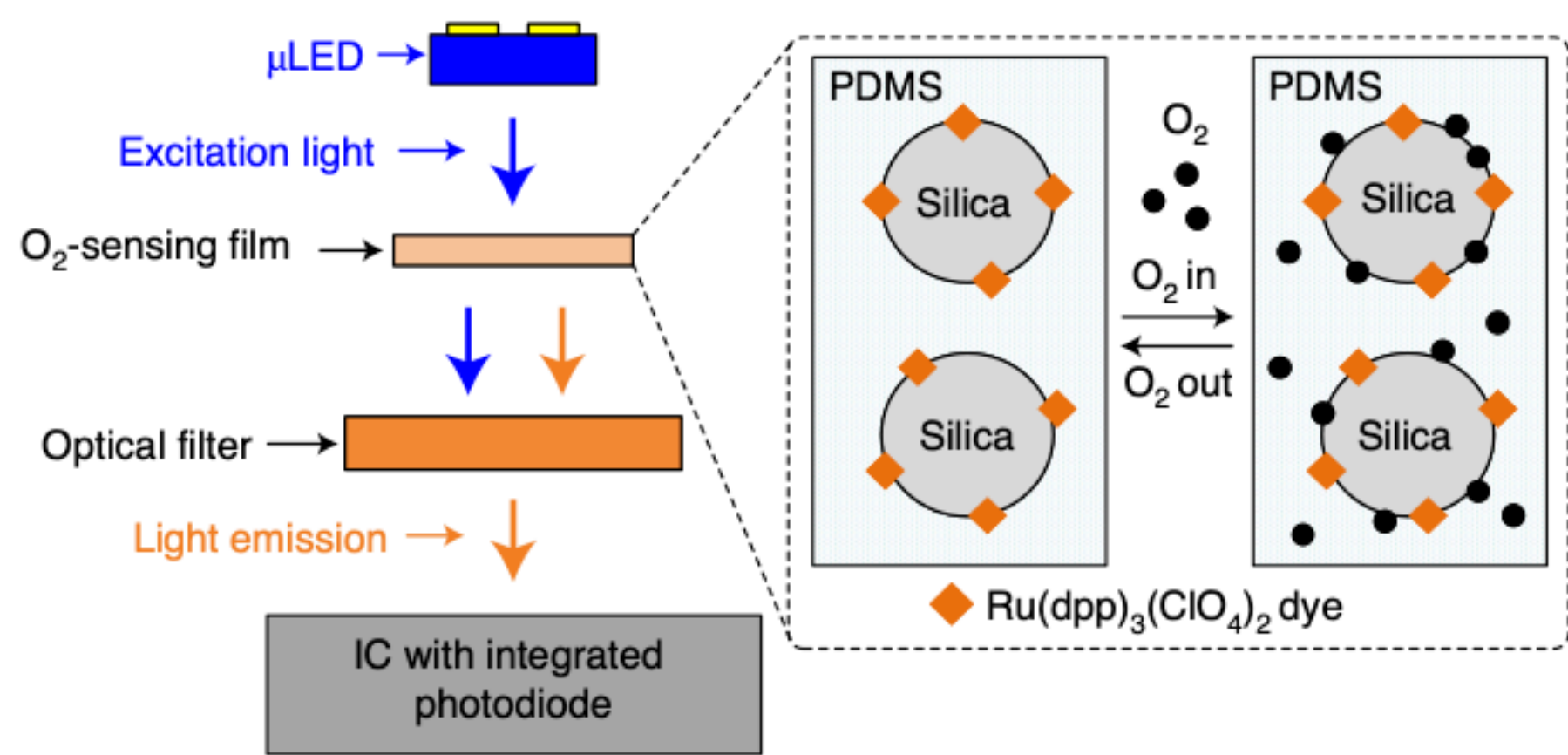
ARTICLES

<https://doi.org/10.1038/s41587-021-00866-y>

Check for updates

Monitoring deep-tissue oxygenation with a millimeter-scale ultrasonic implant

Soner Sonmezoglu¹✉, Jeffrey R. Fineman^{2,3}, Emin Maltepe^{2,3} and Michel M. Maharbiz^{1,4,5}✉



Questions

- Neuroscience: What could we learn from chronic, high-density recordings?
- What are the critical parameters: High density (how high?), distinct brain regions, time-scales?
- Compare with e.g. miniscope (1200 cells over months)
- Other modalities? Pressure, pH, temperature?
- Single unit AP vs LFPs?
- Clinical: Is non (or minimally) invasive really a pre-requisite for practical BMIs?
- Can people envisage BMIs in healthy humans?

